Testing the Drug Main Effect in Aggregated N-of-1 Trials When the Carryover Half-Life Is Unknown: Estimands, Probability Limits, Randomization Tests and Data-Chosen Half-Lives

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Purpose. The rebuilt paper 04 will test the drug main effect β_D in the Hybrid design and its comparators under the refined process, with analyses that weight off-drug visits by an exposure-decayed indicator $D_{bc,it}$ built at an analysis half-life $t^a_{1/2}$ (the working protocol, `analysis/scripts/treatment-main-effect/00-ademp-protocol.md`, here "the protocol"). `docs/57` asked how $t^a_{1/2}$ may be chosen from the data when the target is the biomarker-by-drug interaction. This note asks the same question for the main effect, where the answer is different, and develops the mathematics the protocol's closed-form step needs: what the coefficient of $D_{bc,it}(h)$ estimates under Gompertz onset, expectancy and carryover; its probability limit under a working covariance when the true half-life is $t_{1/2}$ and the analysis uses h; why the half-life cannot be estimated under the null; how size can be controlled when it is chosen anyway (maximum statistics, randomization tests, the bootstrap); what model averaging and joint estimation do to the estimand; and a closed-form power development for the Hybrid design. A worked example on one simulated trial and a modest Monte Carlo check accompany the derivations. A comparison table, a provisional recommendation for paper 04 and a list of proposed edits to the protocol close the note. It is written for trial statisticians and methodologists and assumes the model statements of `docs/56` and the protocol.

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Notation index and glossary

Notation follows the project's canonical table (`analysis/report/NOTATION.md`). Symbols marked *local* are defined here or in the protocol for the main-effect work; Appendix A lists the source of every symbol.

Table 1: Notation index for this note

Symbol	Meaning	Status
i, t	participant; visit, with $t = 0$ the baseline row	canonical
w_t	calendar week of visit t	canonical
N, P	total participants across paths; number of randomization paths	canonical
p, $p(i)$	a path; the path of participant i (the protocol's path letter, allowed by `NOTATION.md`'s q, p row)	local (protocol)
ϖ_p	expected share of path p, $1/P$	local (protocol)
Y_{it}, $\mathbf{Y}_i$	symptom score (code `Sx`); participant i's nine scores	canonical
BL_i, TV_{it}, PB_{it}, BR_{it}	baseline level; natural-course, placebo-belief and drug responses	canonical
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `De`)	canonical
D_{it}, $D_{bc,it}$	binary drug state; exposure-decayed indicator with the pre-exposure guard (code `Dbc`, `X` in the scripts)	canonical
t_{od}, t_{sd}, t_{pb}	time on drug, time since discontinuation, time under expectancy (weeks)	canonical
$t_{1/2}$, $t^a_{1/2}$	true carryover half-life; half-life assumed by an analysis (weeks)	canonical
h	a candidate value of $t^a_{1/2}$; $D_{bc,it}(h)$ is the indicator built with it	local (`docs/57`)
$\mathcal{T}$	the analysis grid $\{0, 0.2, 0.5, 1, 2\}$ weeks	local
r_1	$2^{-1/h}$, the share of the drug effect that $D_{bc}(h)$ keeps one week after stopping	local
m_{BR}, f_m	Gompertz ceiling of BR (published 10.986); drug-effect multiplier $m_{BR}/10.986$	canonical (m_{BR}); local, protocol (f_m)
Ψ_{pt}, Ψ_p	unit drug-response profile of path p at visit t, and its vector over visits: the BR mean divided by m_{BR}, anchored carryover included, 0 at baseline (renamed 2026-10-09 from q_{pt}, $\mathbf{q}_p$, which collided with the canonical path index q)	canonical
v_i, τ^2	participant-specific deviation of the ceiling; its variance (participant-by-treatment variance)	local, protocol (v_i); canonical (τ^2)
β_D	drug main effect	canonical
β_D^{full}, β_D^{exp}, β_D^{bl}	the three candidate estimands of the protocol, Section 4.1	local (protocol)
μ_p, μ^0	mean of $\mathbf{Y}_i$ on path p; its non-drug part, common to all paths	local
Σ_p	covariance of $\mathbf{Y}_i$ on path p	local
$X_p(h)$, $\mathbf{x}_p(h)$	design matrix of an analysis on path p; its drug column $D_{bc}(h)$	local
V	working covariance of an analysis	local
$\omega_p(h)$	weights of the drug coefficient on path p (Section 5.1)	local
$b_0(h)$	null bias of the drug coefficient: its limit when $f_m = 0$	local
$\bar{\Psi}_\omega(h; t_{1/2})$	exposure-weighted drug profile, $\sum_p \varpi_p \omega_p(h)' \Psi_p$ (formerly $\bar{q}_\omega$)	local
$\bar{d}(h)$	on-minus-off mean of $D_{bc}(h)$ over post-baseline visits	local
$\hat{\beta}_D^{\mathrm{exp}}(h)$	model-implied exposure contrast, $\hat{\beta}_D(h)\,\bar{d}(h)$	local
$T(h)$, $\mathrm{LR}(h)$	CR2 t-statistic of the drug coefficient at h; likelihood-ratio statistic for it	local (`docs/57`)

Symbol	Meaning	Status
$\ell_1(h)$, ℓ_0	maximized ML log-likelihood with $D_{bc}(h)$; without any drug term	local
$c_{\max}$	95% critical value of $\max_{h\in\mathcal{T}}\lvert T(h)\rvert$	local
$\widehat{C}$, $\widehat{K}$	the contrast part and the relabeling-invariant part of a linear statistic (Section 9.3)	local
Λ	noncentrality of a test statistic	canonical
σ_u^2, ρ_a	random-intercept variance and CAR(1) parameter of the working covariance	canonical
ρ, c_1	AR(1) serial correlation per week (0.7); cross-component correlation (0.2)	canonical
α, π	nominal level (0.05); power	canonical
M	relabelings or resamples per trial	local
n_{sim}	Monte Carlo replicates per cell (formerly R)	canonical
$U_i(h)$	participant i's contribution to the CR2 score of the drug coefficient at h (formerly $u_i(h)$, which collides with the canonical random intercept u_i)	local
$\Pr(\cdot)$, $\mathbb{E}(\cdot)$	probability and expectation (P is the number of paths)	canonical

Two collisions are avoided deliberately. `docs/57` wrote τ^2 for the random-intercept variance and θ for $\log t_{1/2}^a$ before its 2026-10-09 notation pass; here τ^2 is the participant-by-treatment variance, as in the protocol, the random-intercept variance is σ_u^2, and θ (reserved for a generic interaction coefficient) is not used. The reserved symbols γ, δ, κ, ν, θ and λ do not appear; π is power only.

Sign convention (NOTATION.md, rule 5). The components are non-negative reductions in symptom severity, so β_D is negative when the drug helps, and a positive bias is a bias toward no effect.

Units. Carryover half-lives are in weeks throughout, as in the protocol and, since 2026-10-09, in `NOTATION.md` for paper 04 (1 week = 7 days). Effects are in CAPS points.

Glossary.

- **Designs.** *Hybrid*: open label at weeks 4 and 8, blinded discontinuation and a brief blinded crossover, visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, four paths [1]. *OL+BDC*: open label to week 16, then blinded discontinuation at week 17 or 18, two paths. *CO*: two-period crossover, visits every 2.5 weeks, two paths (protocol, Table "Designs").
- **The refined process.** The recommended data-generating process of paper 01 (Table `tab:spec`): separable AR(1) with $\rho = 0.7$ and $c_1 = 0.2$, anchored carryover, a stop on matrices that are not positive definite, randomized path allocation. **The published process**: Hendrickson's code at commit 58b32a9 (`NOTATION.md`).
- **Anchored and compounding carryover.** Decay of the off-drug drug response from its last on-drug value by the elapsed time, against the published recursion that decays an already decayed value (`docs/53`).
- **Analyses.** All have a random intercept and CAR(1) residuals on weeks (`nlme::lme`, REML) and CR2 inference with Satterthwaite degrees of freedom [3, 4], with the baseline as a response row. **F1**: $Y \sim t + D_{bc}(h) + \eta_t$, the protocol's proposed reference. **S**: $Y \sim t + D_{bc}(h)$. **VF**: visit effects plus $D_{bc}(h)$ (protocol, Section 5.2). **F1A**, **VFA**: the same with h chosen by AIC over $\mathcal{T}$ (ML for the choice, REML refit).
- **Sharp null.** No drug effect for anyone: $f_m = 0$ and $\tau = 0$, so each participant's outcome distribution does not depend on the path. **Weak null.** Zero mean effect with heterogeneity: $f_m = 0$, $\tau > 0$ (the protocol's "heterogeneous null").
- **Working parameter.** A parameter of the analysis model that need not equal any parameter of the data-generating process; here h (`docs/57`, Section 3.2).

Evidence labels. *Quoted* (reproduced from a source), *inspected* (read from code or stored output), *derived* (follows algebraically, to first order where stated), *verified* (computed by running code; scripts in Section 15), *argued* (reasoned, not derived in full or computed). Results labeled verified come from a closed-form computation unless the text says Monte Carlo.

1. Summary for the busy reader

The question. Paper 04 tests β_D with $D_{bc,it}(h)$ in the model. The project wants the half-life chosen from the data (`docs/57`). For the interaction, `docs/57` found that a selection blind to the tested term (F1A0) is approximately harmless. For the main effect no such selection exists: h enters the model only through the tested column.

Five results (Sections 4 to 9).

1. **The coefficient of $D_{bc}(h)$ is a weighted average of the drug profile plus a null bias** (derived): its probability limit is $b_0(h) + \beta_D^{\text{full}}\,\bar{\Psi}_\omega(h; t_{1/2})$, where the weights come from the design, the time adjustment and the working covariance. Its scale changes with h; the model-implied exposure contrast $\hat{\beta}_D(h)\,\bar{d}(h)$ is the quantity comparable across h and with β_D^{exp}.
2. **A fixed half-life does not by itself guarantee size for the main effect.** The null limit $b_0(h)$ is exactly 0 for every h and any working covariance when the time adjustment contains visit effects (VF), or in the CO design (derived). Under F1 it is +0.33 points in the Hybrid design (SE 1.22 at $N = 70$), which gives size 0.058 at $N = 70$ and 0.068 at $N = 150$ by the normal approximation; under S it is −0.61 (size 0.083) in the Hybrid design and −2.23 (size 0.47) in OL+BDC (verified). This corrects the expectation, carried over from the interaction, that any fixed h is right-sized.
3. **AIC choice of h is the maximum-$|t|$ test with a one-test critical value** (derived): because the model without the drug term does not depend on h, $\ell_1(h) = \ell_0 + \text{LR}(h)/2$, so AIC picks the h with the largest likelihood-ratio statistic. Its size is $\Pr(\max_h |T(h)| > 1.96)$: 0.088 for F1 and 0.077 for VF in the Hybrid design, against a calibrated critical value $c_{\max} \approx 2.15$ (verified, closed form); the Monte Carlo check gives 0.093 and 0.087 (MCSE 0.012; Section 11).
4. **A randomization test is exact under the sharp null and can contain the selection,** and it is cheap if the working covariance is taken from the null model, whose design does not depend on the path labels (derived). It uses only the randomized between-path contrasts: at $f_m = 0.4$ its power is 0.39 whatever statistic it is applied to, against 0.63 for the model-based F1 test, whose extra power rests on within-path contrasts protected only by the time and expectancy model (verified, closed form). Under the weak null an unstudentized permutation test is not exact; in the Hybrid design the variance ratio is 1.027 at $\tau = 5.49$ (size 0.053), and with two equally allocated paths it is exactly 1 (derived, verified).
5. **The half-life is barely identified for the main effect at realistic effect sizes.** At $f_m = 0.4$ and $N = 70$ the expected ML deviance of F1 between $h = 0$ and $h = 2$ weeks is below 2 for every true half-life examined; at the published ceiling it reaches 10 to 15 against a short truth (verified). The best-fitting h is a working value that depends on the effect size (Section 9.6).

Anchored against compounding carryover matters more for the main effect than for the interaction: at a true half-life of 1 week the compounding recursion inflates β_D^{exp} by 11% in the Hybrid design and 14% in OL+BDC and F1's power by 0.047 and 0.068 (verified), where the interaction's closed-form power moved by at most 0.002 (quoted, paper 01, Appendix B.3).

The worked example (one trial, $N = 70$, $f_m = 0.4$, true $t_{1/2} = 0.2$, truth $\beta_D^{\text{exp}} = -2.85$; Section 10): F1 at $h = 0.2$ estimated an exposure contrast of −2.61 ($p = 0.031$); paying for a data-chosen h gave $p = 0.047$ (multivariate normal), 0.049 (Davies) and 0.056 (exact randomization with the maximum inside), while every test resting on the randomized contrast alone gave 0.11 to 0.19. The profile could not exclude any half-life. The Monte Carlo check (Section 11) reproduced the closed-form moments, sizes and powers within its error.

Provisional recommendation (Section 12). Keep a half-life fixed in advance (0.2 week) in the primary analysis and add a randomization test of a visit-effects statistic with the grid maximum inside it, as the size-exact companion; report the profile of h and the model-implied exposure contrasts across the grid; do not use F1A or AIC averaging as tests. Everything here is provisional.

2. Scope and relation to other documents

The protocol (Sections 3 to 5) fixes the data-generating process, the designs and the analyses; its Appendix A reviews 38 Monte Carlo and closed-form studies of main-effect tests in N-of-1 and single-case designs, and this note does not repeat that review (Section 13 cross-references it). `docs/56` states F1 for the interaction and its pre-exposure guard; `docs/57` treats the data-chosen half-life for the interaction. The closed form here extends the compendium's separation result (`compendium-mathematics.tex`, eq:bp-plim) and its probability limit of the drug coefficient (eq:bp-fwl; `docs/42`, Section 4) from working independence to the random-intercept CAR(1) working covariance, and adds the decomposition by half-life. All numbers are for the refined process with $c_{bm} = 0$ (no biomarker term), $\rho = 0.7$, anchored carryover and equal expected path shares, unless a table says otherwise. Nothing here is a stress test.

3. Setting

3.1 The data-generating process

For participant i on path p, under the refined process with the protocol's random drug slope (protocol, Sections 3.1 and 3.2),

$$Y_{i0} = \mathrm{BL}_i, \qquad Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}) - v_i\,\Psi_{pt}, \quad t \geq 1,$$

with $\mathbb{E}(BR_{it}) = f_m m_{BR}\,\Psi_{pt}$, so that

$$\mu_p = \mu^0 + \beta_D^{\text{full}}\,\Psi_p, \qquad \beta_D^{\text{full}} = -f_m m_{BR}, \qquad \Sigma_p = \Sigma + \tau^2\,\Psi_p\Psi_p'.$$

$\mu^0 = \mathbb{E}(\mathrm{BL}) - \mathbb{E}(TV) - \mathbb{E}(PB)$ is the non-drug mean. In all three designs it is the same on every path, because the paths share the visit schedule and the expectancy pattern (inspected, the `designs` list of `04-power-simulation.R`). Σ, the covariance of the components without heterogeneity, is also common to all paths, because the BR standard deviation is 8 on and off drug (protocol, Section 3.3). The profile Ψ_p is the Gompertz curve in time on drug, $G(t_{od}; 1, 5, 0.42)$ rescaled to the unit ceiling, restarting at re-exposure, with anchored decay $\Psi_{\text{last}}\,2^{-t_{sd}/t_{1/2}}$ off drug. On Hybrid path 1 at $t_{1/2} = 0.5$ week it is 0, 0.390, 0.839, 0.891, 0.927, 0.232, 0.058, 0.390, 0.002 at weeks 0 to 20 (verified, `58-common.R`).

The sharp null holds when $f_m = 0$ and $\tau = 0$ (derived). Then $\mu_p = \mu^0$ and $\Sigma_p = \Sigma$ for every path, so the distribution of $\mathbf{Y}_i$ does not depend on the path: the outcome vectors are exchangeable across path labels. This is the property a randomization test needs (Section 9.3).

3.2 The analyses and the half-life

An analysis fits $\mathbf{Y}_i = X_{p(i)}(h)\xi + \mathbf{e}_i$ by GLS with a working covariance V, where $X_p(h)$ contains the non-drug columns (intercept, linear time, η_t; or visit indicators) and the drug column $\mathbf{x}_p(h)$:

$$x_{pt}(h) = D_{bc,pt}(h) = \begin{cases} 1 & \text{on drug} \\ 2^{-t_{sd,pt}/h} = r_1^{\,t_{sd,pt}} & \text{off drug after first exposure} \\ 0 & \text{at baseline and before first exposure,} \end{cases}$$

with $h = 0$ giving D_{pt} (the guard of `docs/56`, Section 3.2). In the Hybrid design the off-drug visits all follow exposure, at t_{sd} of 1, 2 and 4 weeks on path 1; 1, 2 and 6 on path 2; 1, 2, 3 and 4 on path 3; 1, 2, 3 and 7 on path 4 (inspected). $D_{bc}(h)$ is therefore a polynomial in $r_1 = 2^{-1/h}$, the share of the effect the indicator keeps one week after stopping: $r_1 = 0.031$ at $h = 0.2$, 0.25 at $h = 0.5$, 0.5 at $h = 1$ and 0.71 at $h = 2$ (derived). Any $h \leq 0.15$ gives $r_1 < 0.01$, so all half-lives in the pharmacological range of the protocol (0.1 to 0.2 week) give nearly the binary indicator at weekly visits.

The analysis grid is $\mathcal{T} = \{0, 0.2, 0.5, 1, 2\}$ weeks: the grid of `docs/57` with 0.25 replaced by the protocol's default $t^a_{1/2} = 0.2$.

4. Estimands

4.1 Definitions and values

The protocol (Section 4.1) defines three candidates on the β_D scale:

$$\beta_D^{\text{full}} = -f_m m_{BR}, \qquad \beta_D^{\text{exp}} = \beta_D^{\text{full}} \frac{1}{P} \sum_p \left(\bar{\Psi}_p^{\text{on}} - \bar{\Psi}_p^{\text{off}} \right), \qquad \beta_D^{\text{bl}} = \text{the same over blinded visits only,}$$

where $\bar{\Psi}_p^{\text{on}}$ and $\bar{\Psi}_p^{\text{off}}$ are means of Ψ_p over the post-baseline on-drug and off-drug visits. The script reproduces the protocol's values at $t_{1/2} = 0$: Hybrid -7.22, OL+BDC -9.35, CO -6.70 for β_D^{exp}, and Hybrid -7.56 for β_D^{bl} (verified against the protocol's `estimand-check.R` values, quoted). Table 2 adds the dependence on the true half-life and on the carryover form.

Table 2: Estimand values at the published ceiling ($f_m = 1$; every value scales with f_m), equal path weights (verified, `58-cf-estimands.csv`)

Design	$t_{1/2}$ (weeks)	β_D^{exp}, anchored	β_D^{exp}, compounding	β_D^{bl}, anchored
Hybrid	0	−7.22	−7.22	−7.56
Hybrid	0.2	−7.13	−7.13	−7.47
Hybrid	0.5	−6.29	−6.44	−6.63
Hybrid	1	−4.82	−5.34	−5.16
Hybrid	2	−2.95	−3.77	−3.29
OL+BDC	0.2	−9.20	−9.20	−10.80
OL+BDC	1	−5.70	−6.47	−7.30
CO	0.2	−6.70	−6.70	−6.70
CO	1	−6.43	−6.47	−6.43

Two readings (derived from the table). In the pharmacological range the estimands barely move (-7.22 to -7.13 in the Hybrid design), because the effect is almost gone one week after stopping. At one week or more carryover fills the off-drug visits and β_D^{exp} shrinks by a third or more, which is a property of the estimand, not a bias.

4.2 What the coefficient of $D_{bc}(h)$ estimates

Proposition 1 (decomposition; derived). Let $B = \sum_p \varpi_p X_p' V^{-1} X_p$ and let $\omega_p(h)' = \mathbf{e}_D' B^{-1} X_p(h)' V^{-1}$ be the row of the GLS projection that gives the drug coefficient on path p. Then $\hat{\beta}_D(h) = N^{-1} \sum_i \omega_{p(i)}(h)' \mathbf{Y}_i + o_p(N^{-1/2})$ with the weights at the limit of the working covariance, and

$$\operatorname{plim} \hat{\beta}_D(h) = \sum_p \varpi_p \, \omega_p(h)' \mu_p = \underbrace{\sum_p \varpi_p \, \omega_p(h)' \mu^0}_{b_0(h)} + \beta_D^{\text{full}} \underbrace{\sum_p \varpi_p \, \omega_p(h)' \Psi_p}_{\bar{\Psi}_\omega(h;\, t_{1/2})}.$$

The weights reproduce the columns: $\sum_p \varpi_p \omega_p' \mathbf{x}_p(h) = 1$ and $\sum_p \varpi_p \omega_p' \mathbf{z}_p = 0$ for every other column $\mathbf{z}_p$ (derived, from $B^{-1}B = I$). This is the compendium's eq:bp-fwl generalized to any working covariance and any half-life; at working independence with visit effects it reduces to that equation.

Three consequences (derived):

- **The coefficient is a weighted average of the drug profile, on the scale of $D_{bc}(h)$.** If the true profile lay in the span of $\mathbf{x}_p(h)$ and the non-drug columns, $\bar{\Psi}_\omega = 1$ and the coefficient would be β_D^{full}. Under Gompertz onset the on-drug profile varies from 0.39 to 0.93 across the Hybrid visits, so it never does, and $\bar{\Psi}_\omega$ depends on which visits the analysis weights.
- **The scale changes with h.** A larger h codes the off-drug visits as partly exposed, so the coefficient is the effect "per unit of $D_{bc}(h)$" and grows in magnitude as h grows, even when the data do not change (Table 4, column "plim").
- **The comparable quantity is the model-implied exposure contrast.** The fitted drug part of the mean is $\hat{\beta}_D(h)\, x_{pt}(h)$; its on-minus-off mean over the design's visits, the definition of β_D^{exp} applied to the working model, is

  $$\hat{\beta}_D^{\text{exp}}(h) = \hat{\beta}_D(h)\, \bar{d}(h), \qquad \bar{d}(h) = 1 - \frac{1}{P} \sum_p \bar{x}_p^{\text{off}}(h),$$

  because $x = 1$ on drug. $\bar{d}(h)$ is 1 at $h = 0$, 0.991 at 0.2, 0.906 at 0.5, 0.755 at 1 and 0.551 at 2 weeks in the Hybrid design (verified). Any comparison or average across h should be made on this scale (Section 9.4).

4.3 Sharp and weak nulls

Every estimand is zero when $f_m = 0$, so size refers to one hypothesis in every design (protocol, Section 4.1). Two nulls must nevertheless be distinguished, as in the literature on series of N-of-1 trials [43, 44] and on randomization inference [28, 29]:

- **Sharp null**, $f_m = 0$ and $\tau = 0$: no participant has a drug effect. The outcome vectors are exchangeable across paths (Section 3.1), so a randomization test is exact (Section 9.3).
- **Weak null**, $f_m = 0$ and $\tau > 0$: individual effects $-v_i \Psi_p$ average to zero. The means are still common to all paths, but the covariances $\Sigma_p = \Sigma + \tau^2 \Psi_p \Psi_p'$ differ by path, so exchangeability fails. A model-based or CR2 test of the mean is still asymptotically valid if its standard error includes the τ^2 term; CR2 does, because it is computed from participant residuals (derived from the sandwich form, `docs/56`, Section 3.5). A randomization test is then valid only asymptotically and only for a suitably studentized statistic (Section 9.3).

5. The probability limit of the drug coefficient

5.1 The limit under a working covariance

Proposition 2 (limit and variance; derived). Under randomized allocation with shares ϖ_p, participants' contributions $X_{p(i)}' V^{-1} (\mathbf{Y}_i - X_{p(i)} \xi^*)$ are independent with mean zero at the limit ξ^*, so

$$\operatorname{Var} \hat{\beta}_D(h) \approx \frac{1}{N} \sum_p \varpi_p \, \omega_p(h)' (\Sigma_p + \mathbf{r}_p \mathbf{r}_p') \omega_p(h), \qquad \mathbf{r}_p = \mu_p - X_p(h) \xi^*,$$

the sandwich with the mean misfit $\mathbf{r}_p$ included; under fixed path counts the $\mathbf{r}_p\mathbf{r}_p'$ term drops. CR2 estimates the first form. The working covariance itself converges to the random-intercept CAR(1) member closest to the truth in Kullback-Leibler distance, which the script finds by minimizing $\sum_p \varpi_p[\log|V|+\mathrm{tr}\{V^{-1}(\Sigma_p+\mathbf{r}_p\mathbf{r}_p')\}]$, the expected ML criterion per participant, with ξ^* recomputed for each candidate V (derived; REML has the same limit). In the Hybrid design under the null the limit has $\rho_a = 0.646$ per week for F1 (verified).

The limit of F1 under the true covariance is close to that under the working covariance, and the working-independence limit of `docs/42` is not: F1's null bias is +0.29 under independence, +0.18 under the true covariance and +0.33 under the random-intercept CAR(1) limit, with standard errors 1.63, 1.21 and 1.22 at $N = 70$ (verified, `58-cf-null.csv`). The working covariance, not only the time adjustment, decides the weights.

5.2 Under the null: when a fixed half-life keeps its size

Proposition 3 (null limit; derived). If the non-drug mean μ^0 lies in the span of the non-drug columns, and those columns are the same on every path, then $b_0(h) = 0$ for every h and every working covariance. Proof: $\mu^0 = Z\mathbf{a}$ for the common non-drug block Z, and $\sum_p \varpi_p\omega_p'Z = 0$ by Proposition 1. Visit effects satisfy the condition in every design (they span all of $\mathbb{R}^9$ on the common schedule). Linear time with or without η_t does not, because the non-drug mean is a sigmoid natural course plus an expectancy curve that halves at week 9 (`docs/42`, Section 2).

So, unlike the interaction, whose null limit is zero for any fixed h because the biomarker block separates (compendium, eq:bp-plim), the main effect is protected for any fixed h only by the time adjustment or by the design's symmetry. Table 3 gives the size this implies.

Table 3: Null bias $b_0(h)$ (points), standard error and normal-approximation size at $N = 70$, random-intercept CAR(1) working covariance at its limit, $t_{1/2} = 0.2$ (verified, `58-cf-null.csv`)

Design	Analysis	b_0, $h = 0$	b_0, $h = 0.2$	b_0, $h = 1$	b_0, $h = 2$	SE, $h = 0.2$	Size, $h = 0.2$
Hybrid	S	−0.58	−0.61	−1.12	−1.63	1.14	0.083
Hybrid	F1	+0.32	+0.33	+0.25	+0.13	1.22	0.058
Hybrid	VF	0	0	0	0	1.66	0.050
OL+BDC	S	−2.16	−2.23	−3.29	−3.92	1.18	0.473
OL+BDC	F1	−0.22	−0.22	−0.09	+0.60	1.42	0.053
OL+BDC	VF	0	0	0	0	2.35	0.050
CO	S	−0.26	−0.26	−0.33	−0.43	1.48	0.053
CO	F1	0	0	−0.06	−0.15	1.59	0.050
CO	VF	0	0	0	0	1.59	0.050

Readings (verified unless stated):

- **F1 is nearly, not exactly, right-sized in the Hybrid design.** Its null bias of +0.33 points is a quarter of a standard error at $N = 70$; because the bias is fixed and the standard error falls as $N^{-1/2}$, the size grows with N: 0.054 at $N = 35$, 0.058 at 70, 0.068 at 150 and 0.086 at 300 (derived from Table 3 by scaling the standard error). The protocol's size screen (pooled rate within 2 MCSE of 0.05, no cell above 0.075) would pass F1 at $N \leq 150$ and fail it at large N.
- **S, linear time without η_t, is not usable for the main effect in OL+BDC** and is liberal in the Hybrid design, as the protocol predicted from `docs/40` (protocol, Section 5.1).
- **The null bias depends on** h under F1 and S, so a data-chosen h selects among biased statistics (Section 9.1).
- **VF and CO are protected exactly.** In CO the two paths are mirror images and F1's columns absorb what remains; the small b_0 at $h \geq 1$ comes from the asymmetric guard on the placebo-first path.

5.3 Under the alternative: bias and power as h departs from $t_{1/2}$

Table 4 gives the limit, the model-implied exposure contrast and the power of F1 and VF in the Hybrid design at $f_m = 0.4$ ($m_{BR} = 4.39$ points), for true and analysis half-lives on the grid.

Table 4: Hybrid design, $f_m = 0.4$, $N = 70$: probability limit of the drug coefficient and of the model-implied exposure contrast, standard error and normal-approximation power for F1 and VF (verified, `58-cf-alt.csv`; all grid values in the file)

True $t_{1/2}$	h	F1 plim	F1 $\widehat{\beta}_D^{\text{exp}}$ limit	β_D^{exp}	F1 SE	F1 power	VF $\widehat{\beta}_D^{\text{exp}}$ limit	VF SE	VF power
0	0	−2.81	−2.81	−2.89	1.20	0.651	−2.88	1.62	0.428
0	1	−3.26	−2.46	−2.89	1.60	0.533	−2.18	2.21	0.257
0.2	0	−2.74	−2.74	−2.85	1.20	0.630	−2.80	1.62	0.407
0.2	0.2	−2.78	−2.76	−2.85	1.22	0.627	−2.81	1.66	0.400
0.2	1	−3.23	−2.44	−2.85	1.60	0.526	−2.14	2.21	0.250
0.2	2	−3.62	−2.00	−2.85	1.96	0.455	−1.62	2.71	0.192
0.5	0	−2.24	−2.24	−2.52	1.20	0.466	−2.22	1.62	0.278
0.5	0.5	−2.59	−2.35	−2.52	1.37	0.473	−2.24	1.92	0.251
1	0	−1.59	−1.59	−1.93	1.20	0.263	−1.63	1.62	0.171
1	1	−2.45	−1.85	−1.93	1.59	0.336	−1.59	2.21	0.158
1	2	−3.10	−1.71	−1.93	1.96	0.353	−1.32	2.71	0.142
2	0	−0.91	−0.91	−1.18	1.20	0.118	−1.08	1.62	0.102
2	2	−2.41	−1.33	−1.18	1.96	0.234	−0.98	2.71	0.101

Readings (verified unless stated):

- **In the pharmacological range the half-life hardly matters.** At true $t_{1/2} = 0.2$, every $h \leq 0.5$ gives F1 power between 0.590 and 0.630; a wrong $h = 1$ costs 0.10 and $h = 2$ costs 0.17, because a long h codes the off-drug visits as exposed and inflates the standard error (1.22 to 1.96).
- **Under long carryover the best h is longer than the truth.** At true $t_{1/2} = 1$, F1's power is 0.263 at $h = 0$, 0.336 at $h = 1$ and 0.353 at $h = 2$. Power peaks near the working value that best fits the misspecified mean, not at the pharmacological half-life (the working-parameter point of `docs/57`, Section 3.2; Section 9.6).
- **F1 targets β_D^{exp} closely when h is near the truth**: its exposure contrast limit is within 0.09 point (3%) of β_D^{exp} at $t_{1/2} = h = 0.2$, including the null bias, and within 0.17 at $t_{1/2} = h = 0.5$. A wrong long h attenuates it by 14% to 30%.
- **VF is unbiased under the null but pays in precision** (SE 1.66 against 1.22, power 0.40 against 0.63) and targets the randomized contrast, whose value is close to β_D^{exp} in the Hybrid design and equal to β_D^{bl} in OL+BDC (−4.38 at $f_m = 0.4$; verified).

In OL+BDC F1's exposure contrast overstates β_D^{exp} by 21% at $t_{1/2} = h = 0.2$ (−4.40 against −3.68), because the four open-label visits carry the rising Gompertz response at full expectancy and a linear time term with η_t cannot separate the two; in CO the limit is within 10% of the estimand at every h and true half-life (verified, `58-cf-alt.csv`).

6. Expectancy confounding and the half-life

The open-label visits are all on drug at $\eta_t = 1$, the blinded visits have $\eta_t = 0.5$ and baseline has neither. Over the nine rows with equal path weights the correlation of η_t with D_{it} is 0.588 (0.591 with $D_{bc}(0.2)$; 0.509 over the eight post-baseline rows), reproducing the compendium's 0.588 (verified; compendium, Section "The published Hybrid construct").

How F1 identifies β_D (derived from Proposition 1). F1's weights are orthogonal to the intercept, linear time and η_t columns. The open-label rows carry the only $\eta_t = 1$ values, so their contribution to the drug coefficient is limited to what the linear time and η_t restrictions leave over. Table 5 splits the weights by row group.

Table 5: Weights of the drug coefficient by row group (Hybrid, $h = t_{1/2} = 0.2$, random-intercept CAR(1) limit), and F1's null bias and drug signal by row group; the null mean is centered, which changes no total (verified, `58-cf-rowgroups.csv`)

Row group	S: share of $\sum\lvert\omega\rvert$	F1: share	VF: share	F1: contribution to b_0	F1: contribution to $\bar{\Psi}_\omega$ times m_{BR}
baseline	0.104	0.030	0.005	−0.52	0
open label, on drug	0.149	0.097	0.007	+0.61	+1.03
blinded, on drug	0.351	0.466	0.488	−1.21	−8.96
blinded, off drug	0.396	0.406	0.501	+1.44	+0.17
total	1	1	1	+0.33	−7.76

Readings (derived from the table, verified numbers):

- **Adding η_t moves the weight to the blinded rows**, from 75% under S to 87% under F1; VF puts 99% there. The blinded on-off contrast at matched expectancy is what identifies β_D once the expectancy step is modeled, as the protocol's estimand discussion intends (protocol, Section 4.1).
- **The open-label rows enter F1 with net negative weight** and reduce the drug signal by 1.03 points per unit ceiling: F1 uses them mainly to estimate the time and expectancy terms, and the residual Gompertz rise between weeks 4 and 8 then feeds back into the drug coefficient.
- **The null bias is a sum of four non-small parts that nearly cancel** (−0.52, +0.61, −1.21, +1.44). Its size is therefore fragile to the shape of the natural course and the placebo curve, which a real trial does not know (argued).

The between-path information. At weeks 10, 16 and 20 the paths differ in drug state; under $h > 0$ they also differ at weeks 11 and 12, where paths 1 and 2 are one week from stopping and paths 3 and 4 two weeks (inspected). With visit effects only these between-path differences identify β_D, and the plim depends only on the between-path differences of the drug profile: $\operatorname{plim} \hat{\beta}_D = \beta_D^{\text{full}} \sum_p \varpi_p \omega_p' (\Psi_p - \bar{\Psi})$, because $\sum_p \varpi_p \omega_p = 0$ under visit effects (derived). A positive h therefore lets VF borrow weeks 11 and 12, at the price of assuming the carryover shape there; this is how h interacts with the expectancy protection (argued). The randomization test of Section 9.3 uses exactly this between-path information whatever statistic it is applied to.

7. Anchored against compounding carryover

Why the form matters more for the main effect (derived). For the main effect the off-drug means are part of the signal: by Proposition 1 the limit is $\sum_p \varpi_p \omega_p' \mu_p$, and the compounding recursion of the published process lowers the off-drug drug response from the second off-drug visit of each run onward (`docs/53`, Section 1), which raises the on-minus-off contrast directly. For the interaction the slope depends on the coupling g_t, not on the mean, and the carryover of the mean enters only through the path-mean term of the pooled contrast, under 0.7% of the residual variance in the Hybrid design (quoted, `docs/37`, Section 4.4); paper 01 found the closed-form power changed by at most 0.002 (quoted, Appendix B.3).

Table 6: Anchored against compounding carryover at $f_m = 0.4$, $N = 70$, $h = t_{1/2}$ (verified, `58-cf-anchored-recursive.csv`)

Design	$t_{1/2}$	β_D^{exp}, anchored	β_D^{exp}, compounding	F1 power, anchored	F1 power, compounding	VF power, anchored	VF power, compounding
Hybrid	0.2	−2.85	−2.85	0.627	0.627	0.400	0.400
Hybrid	0.5	−2.52	−2.58	0.473	0.487	0.251	0.249
Hybrid	1	−1.93	−2.14	0.336	0.383	0.158	0.158
Hybrid	2	−1.18	−1.51	0.234	0.302	0.101	0.112
OL+BDC	1	−2.28	−2.59	0.490	0.558	0.129	0.125
OL+BDC	2	−1.28	−1.71	0.180	0.258	0.075	0.075
CO	1	−2.57	−2.59	0.329	0.332	0.335	0.338

Readings (verified): in the pharmacological range the two forms agree to three decimals. At $t_{1/2} = 1$ week the compounding form inflates β_D^{exp} by 11% (Hybrid) and 14% (OL+BDC) and F1's power by 0.047 and 0.068; at 2 weeks by 28% and 34%, and 0.068 and 0.078. VF changes little, because the second off-drug visits are at weeks 11 and 12, where VF's weights are smaller. CO has a single off-drug run of four visits 2.5 weeks apart and is barely affected. Any main-effect result from the stored stress test, which used the compounding form, should not be carried into paper 04 at $t_{1/2} \geq 1$ week (argued).

8. Closed-form power for the main effect

Proposition 4 (power; derived). For a linear analysis with weights $\omega_p(h)$, by Propositions 1 and 2 the statistic $\hat{\beta}_D(h)/\text{SE}$ is approximately normal with noncentrality

$$\Lambda(h) = \sqrt{N}\,\frac{b_0(h) + \beta_D^{\text{full}}\,\bar{\Psi}_\omega(h; t_{1/2})}{\left[\sum_p \varpi_p \omega_p'(\Sigma + \mathbf{r}_p\mathbf{r}_p')\omega_p + \tau^2 \sum_p \varpi_p (\omega_p'\Psi_p)^2\right]^{1/2}}, \qquad \pi \approx \Phi(\Lambda - z_{0.975}) + \Phi(-\Lambda - z_{0.975}).$$

This is the main-effect analog of `docs/37`'s E9 development: E9 is a fixed per-participant contrast $\mathbf{a}$ of the outcome regressed on the biomarker; here the contrast is ω_p, its mean is the exposure-weighted drug profile instead of the coupling gap, and heterogeneity enters as $\tau^2(\omega_p'\Psi_p)^2$, which is the protocol's $\tau^2(\mathbf{a}'\Psi)^2$ (protocol, Section 5.1). For VF under working independence, ω_p is the FWL contrast $D_{pt} - \bar{D}_t$ of the compendium's eq:bp-fwl (the protocol's summary measure PR), and the moments are exact given the path counts. The normal reference is adequate here: the CR2 Satterthwaite degrees of freedom for the drug coefficient were 64 to 69 at $N = 70$ in the worked example (Section 10), against 21 to 30 for the interaction (`docs/56`, Section 3.5), because every participant contributes an exposure contrast, not only those with extreme biomarker values (verified for the example; argued in general).

Table 7: Normal-approximation power at $f_m = 0.4$, $h = t_{1/2} = 0.2$, by design, analysis, serial correlation, heterogeneity and N (verified, `58-cf-power-grid.csv`, which also has $\tau = 2.75$ and every ρ)

Design	Analysis	ρ	$\tau = 0$, $N = 35$	70	150	$\tau = 5.49$, $N = 70$	SE ($\tau = 0$, $N = 70$)
Hybrid	F1	0.5	0.248	0.439	0.753	0.408	1.48
Hybrid	F1	0.7	0.365	0.627	0.917	0.568	1.22
Hybrid	F1	0.8	0.507	0.798	0.983	0.717	1.02
Hybrid	VF	0.7	0.226	0.400	0.704	0.379	1.66
OL+BDC	F1	0.7	0.602	0.881	0.996	0.817	1.42
OL+BDC	VF	0.7	0.261	0.462	0.779	0.435	2.35

Design	Analysis	ρ	$\tau = 0$, $N = 35$	70	150	$\tau = 5.49$, $N = 70$	SE ($\tau = 0$, $N = 70$)
CO	F1	0.7	0.204	0.358	0.646	0.340	1.59
CO	VF	0.7	0.205	0.360	0.650	0.342	1.59

Readings (verified):

- **Power rises with** ρ in the N-of-1 designs (Hybrid F1 0.44, 0.63, 0.80 at $\rho = 0.5$, 0.7, 0.8), because a stronger serial correlation makes nearby on and off visits more alike apart from the drug, so within-participant contrasts are less noisy.
- **Heterogeneity costs little at these values**: $\tau = 5.49$ points (half the published ceiling) lowers Hybrid F1 power from 0.627 to 0.568 at $N = 70$, because $\omega_p' \Psi_p$ is about 0.7, so it adds $0.5\,\tau^2$ to a contrast variance of about 100.
- **OL+BDC F1's high power is partly its bias** away from zero (Section 5.3), and CO's F1 and VF coincide because its time terms are almost aliased with the visit effects.
- **The half-life enters through** $\bar{\Psi}_\omega$ **and the SE only** (Table 4). With h fixed the closed form is a function of h, $t_{1/2}$, ρ, τ^2 and N, all tabulated in the scripts' outputs.

Figure 1 summarizes Sections 5 to 8 and Section 9.3.

9. Choosing the half-life from the data

9.1 The half-life is not identified under the null

When $\beta_D = 0$ the model $\mathbf{Y}_i = X_p(h)\xi + \mathbf{e}_i$ does not depend on h, because h enters only through the column whose coefficient is zero. The half-life is a nuisance parameter present only under the alternative, the problem of Davies [19, 20], Andrews and Ploberger [21] and Hansen [22]: the score and likelihood-ratio statistics for β_D are functions of h, and their null distributions are those of a process indexed by h, not of a single χ^2_1 variable.

Proposition 5 (AIC selection with the tested term; derived). Every candidate model on $\mathcal{T}$ has the same parameters, so AIC differences are log-likelihood differences. With ℓ_0 the maximized ML log-likelihood of the model without any drug term,

$$\ell_1(h) = \ell_0 + \tfrac{1}{2}\,\mathrm{LR}(h), \qquad \hat{h}_{\mathrm{AIC}} = \arg\max_{h \in \mathcal{T}} \mathrm{LR}(h),$$

and ℓ_0 does not depend on h. With the working covariance held fixed, $\mathrm{LR}(h) = \hat{\beta}_D(h)^2 / \widehat{\mathrm{Var}}_{\mathrm{mb}}\{\hat{\beta}_D(h)\}$ exactly, the square of the model-based Wald statistic; with the covariance re-estimated it holds to first order. F1A therefore selects the h with the largest model-based $|t|$ and refers the CR2 t at that h to a single-test critical value: to first order it is the max-$|t|$ test of Section 9.2 with critical value $t_{0.975}$, and its size is $\Pr(\max_h |T(h)| > t_{0.975}) > \alpha$.

This is the "flat ℓ_0" limiting case of `docs/57`, Section 3.4, holding exactly rather than approximately. The remedy of `docs/57`, choosing h on the model without the tested term (F1A0), has no analog: removing D_{bc} removes every trace of h, so any likelihood-based choice of h must look at the tested term. A data-chosen half-life for the main effect can be valid only if the test is calibrated for the choice (Sections 9.2, 9.3, 9.5), if h is chosen on data the test does not use (sample splitting, `docs/57`, Section 4.3), or if the choice does not use the outcome (pharmacology, fixed in advance).

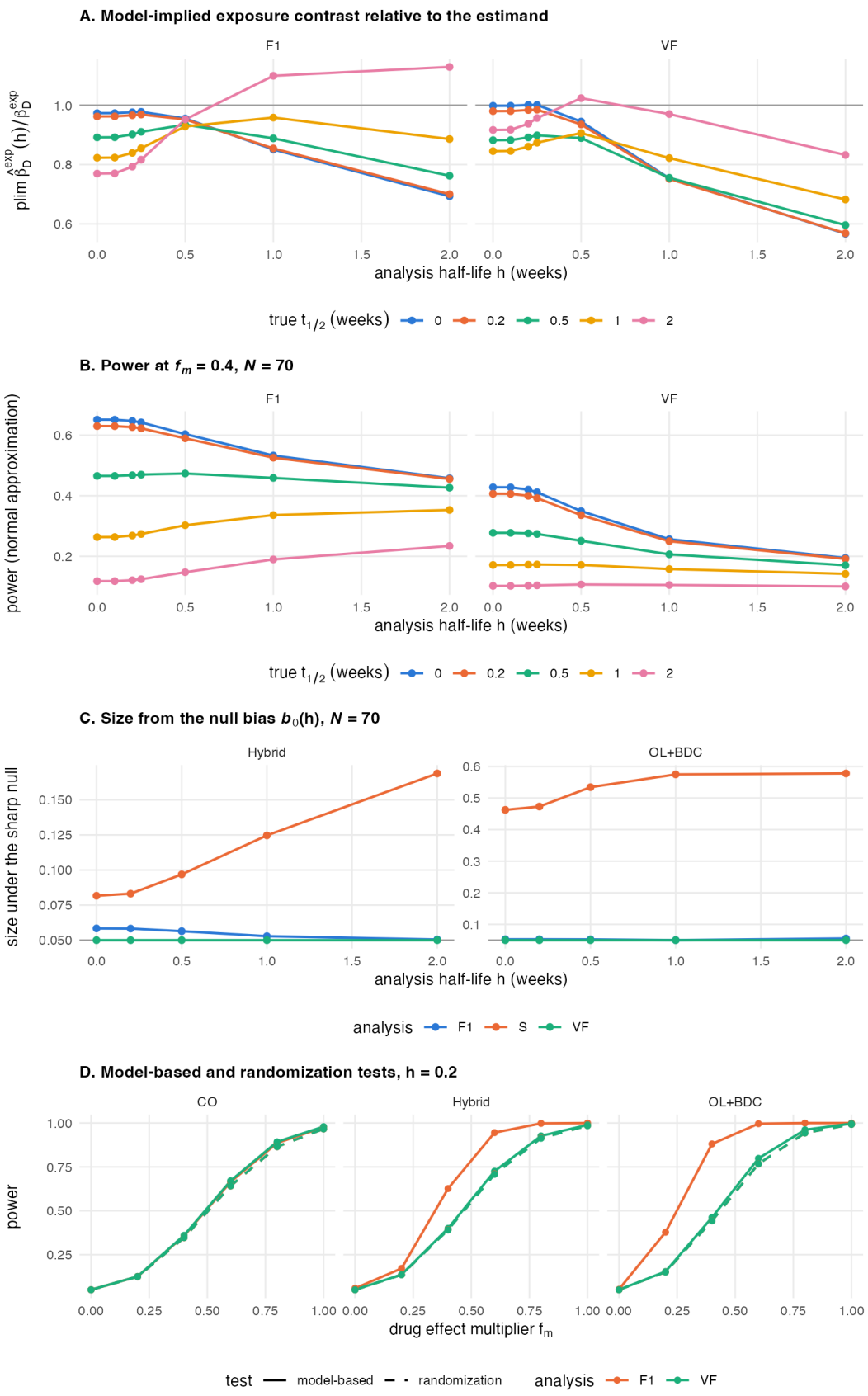


Figure 1: Closed-form results, refined process, $N = 70$. A: limit of the model-implied exposure contrast relative to β_D^{exp} as the analysis half-life h varies, for five true half-lives, F1 and VF, Hybrid, $f_m = 0.4$. B: normal-approximation power in the same cells. C: size under the sharp null implied by the null bias $b_0(h)$, by analysis, Hybrid and OL+BDC. D: power of the model-based test and of the randomization test (Section 9.3) as the drug effect grows, by design, F1 and VF at $h = 0.2$.

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9.2 The maximum statistic over the grid

Joint null distribution (derived). With the working covariance at its limit for each h, the estimates are asymptotically jointly normal with

$$\mathrm{Cov}\{\hat{\beta}_D(h), \hat{\beta}_D(h')\} \approx \frac{1}{N} \sum_p \varpi_p \, \omega_p(h)' (\Sigma_p + \mathbf{r}_p(h)\mathbf{r}_p(h')') \omega_p(h'),$$

so the correlation of $T(h)$ and $T(h')$ is determined by the overlap of the weight vectors, that is by how similar the regressors $D_{bc}(h)$ and $D_{bc}(h')$ are after the other columns are partialled out, measured in the metric of the true covariance. In a trial the same quantity is estimated from the participants' influence contributions $U_i(h) = \mathbf{e}_D' \hat{B}_h^{-1} X_{p(i)}(h)' \hat{V}_h^{-1} \hat{\mathbf{e}}_i(h)$, as $\sum_i U_i(h) U_i(h')$, the cluster-robust covariance of the stacked estimates (the multiplier construction of Hansen [22]). The level-α critical value is the $c_{\max}$ with $\Pr(\max_h |Z_h| \le c_{\max}) = 1 - \alpha$ for Z multivariate normal with that correlation (`mvtnorm::qmvnorm`); a multivariate t with the smallest CR2 degrees of freedom is the small-sample variant.

Table 8: Joint null distribution of $T(h)$ over $\mathcal{T}$, $N = 70$, $t_{1/2} = 0.2$: correlation of the two ends of the grid (the smallest), the 95% critical value of $\max |T|$, and the normal-approximation size of three rules, including the null bias $b_0(h)$ (verified, closed form, `58-tg-maxt-summary.csv`; the multivariate normal integrals are quasi-Monte Carlo, with an error of about 0.005 in $c_{\max}$)

Design	Analysis	$\mathrm{cor}\{T(0), T(2)\}$	$c_{\max}$	Size of F1A-type selection at 1.96	Size at $c_{\max}$ with the null bias	Size at fixed $h = 0.2$
Hybrid	F1	0.827	2.16	0.088	0.056	0.058
Hybrid	VF	0.848	2.14	0.077	0.051	0.050
Hybrid	S	0.867	2.14	0.185	0.138	0.083
OL+BDC	F1	0.783	2.19	0.095	0.057	0.053
OL+BDC	VF	0.811	2.17	0.081	0.050	0.050
CO	F1	0.976	2.04	0.061	0.051	0.050
CO	VF	0.991	2.01	0.056	0.050	0.050

Readings (verified):

- **The five statistics are highly correlated** (0.83 or more in the Hybrid design, 0.98 in CO), so the maximum costs little: $c_{\max}$ is 2.14 to 2.19 rather than the Bonferroni 2.58. In the Hybrid design the F1 correlations fall from 0.999 between $h = 0$ and 0.2 to 0.83 between 0 and 2 weeks.
- **Selection by AIC with a one-test critical value is liberal**: 0.088 for F1 and 0.077 for VF in the Hybrid design. Without the null bias, F1's figure would be 0.080; the selection, not the bias, is most of the excess.
- **Calibration repairs the selection but not the bias**: F1's calibrated max-$|t|$ test has size 0.056 in the Hybrid design, the same as F1 at a fixed h, because the null bias of Section 5.2 remains. VF's calibrated test is at 0.051 (the residual 0.001 is the normal approximation of a maximum over five correlated statistics).

Davies' bound for a continuous range (derived; formula as implemented in `segmented::davies.test`, inspected, citing Davies [20]). For a unit-variance Gaussian process $Z(h)$ the upcrossing argument gives $\Pr(\sup_h Z(h) > u) \le \Phi(-u) + \mathcal{V} e^{-u^2/2}/\sqrt{8\pi}$, with $\mathcal{V} = \sum_j |Z(h_j) - Z(h_{j-1})|$ the observed total variation on a fine grid; for $|Z|$ the union of the two tails gives $2\Phi(-u) + \mathcal{V} e^{-u^2/2}/\sqrt{2\pi}$. The expected value of the observed variation is $\sqrt{2/\pi}$ times the arc length $\sum_j \arccos \mathrm{cor}\{Z(h_j), Z(h_{j-1})\}$, which turns the bound into $2\Phi(-u) + (\text{arc length})\, e^{-u^2/2}/\pi$ (derived). Over $h \in [0, 2]$ weeks the arc length is 0.65 for F1 and 0.59 for VF

in the Hybrid design; the bound puts the critical value at 2.168 and 2.151, against 2.165 and 2.150 from the multivariate normal on the fine grid, and at 2.22 and 2.17 over $[0, 4]$ weeks (verified, `58-tg-davies.csv`). The bound is nearly exact here because the process is smooth and short, and it is a convenient one-line calibration for a continuous h.

Alternatives to the supremum. Andrews and Ploberger [21] derive statistics optimal for weighted average power over the nuisance parameter, the average and the exponential average of the LR process; with five highly correlated statistics the supremum and the averages should differ little (argued), and the supremum is the one whose rejection region a reader can see.

What the maximum test reports. A p-value against the null of no drug effect for any exposure regressor on the grid, and the h at which $|T|$ is largest. It does not provide an estimate with known coverage; Section 9.7 gives intervals that account for the maximum.

9.3 Randomization tests over path assignments

Exactness under the sharp null (derived; [23, 40]). Let the statistic be any function $\mathcal{S}(\mathbf{Y}, \mathbf{L})$ of the outcome vectors and the path labels, including the whole select-then-test procedure (F1A, the maximum over $\mathcal{T}$, AIC averaging). Under the sharp null the outcome vectors are exchangeable across labels (Section 3.1). If $\mathbf{L}$ was drawn from a randomization distribution G and $\mathbf{L}^*$ is drawn from G independently, then $\mathcal{S}(\mathbf{Y}, \mathbf{L})$ and $\mathcal{S}(\mathbf{Y}, \mathbf{L}^*)$ have the same distribution given $\mathbf{Y}$, and the p-value $(1 + \#\{m : \mathcal{S}(\mathbf{Y}, \mathbf{L}^*_m) \geq \mathcal{S}(\mathbf{Y}, \mathbf{L})\})/(M + 1)$ is super-uniform for any number M of relabelings. Nothing about the working covariance, the time adjustment, the half-life or the carryover form enters the argument.

Which randomization distribution. The reference set must be the design's own randomization ("analyze as you randomize"; Rosenberger, Uschner and Wang [41]).

- *Fixed path counts* (the published allocation, 18, 18, 17, 17): uniform over permutations of the labels.
- *The refined process's allocation* (each participant to a path uniformly at random, redrawn if a path receives fewer than 2): either re-draw labels by the same rule, which is exact unconditionally, or permute the observed labels, which is exact conditionally on the realized counts and hence also unconditionally (derived; the counts are ancillary under the sharp null). The conditional version is preferable because the precision of every statistic depends on the counts.
- *Stratified or blocked allocation*: permute within strata or blocks only.

A cheap exact version (derived). Refitting the mixed model for every relabeling costs about 1 second per relabeling with the grid (Section 10). It is avoidable. Fit the null model $Y \sim t + \eta_t$ once; its design is the same on every path, so its estimated covariance $\hat{V}_0$ does not depend on the labels. A GLS statistic computed with $\hat{V}_0$ is then a fixed function of $(\mathbf{Y}, \mathbf{L})$, and with fixed counts each relabeling changes only the path sums $\sum_{i \in p} \mathbf{Y}_i$: B depends on the counts alone. The studentization by a CR0 sandwich with $\hat{V}_0$ is equally cheap. The script `58-common.R` (`perm_setup`, `perm_stats`) computes 999 relabelings of ten statistics in about 2 seconds. The weights are those of the null model rather than of the fitted alternative, which changes the estimate little (-2.699 against -2.700 for F1 at $h = 0$ in the pilot trial; verified).

What a randomization test can use (Proposition 6; derived). Write a linear statistic with label-invariant weights as

$$\hat{\beta} = \underbrace{\frac{1}{N} \sum_i (\omega_{p(i)} - \bar{\omega})' \mathbf{Y}_i}_{\hat{C}} + \underbrace{\bar{\omega}' \frac{1}{N} \sum_i \mathbf{Y}_i}_{\hat{K}}, \qquad \bar{\omega} = \sum_p \frac{N_p}{N} \omega_p.$$

Under permutation with fixed counts $\hat{K}$ does not change, so the test is a test of $\hat{C}$ alone, whose mean is $\beta_D^{\text{full}} \sum_p \varpi_p (\omega_p - \bar{\omega})'(\Psi_p - \bar{\Psi})$: only the between-path differences of the drug profile carry signal. For VF $\bar{\omega} = 0$

and $\hat{K} = 0$. For F1, $\hat{K}$ carries the within-path, time-modeled part of the estimate, including the null bias b_0: at $f_m = 0.4$ in the Hybrid design 45% of F1's limit is in $\hat{K}$ (verified, `58-cf-randomization-power.csv`).

Two consequences (verified, closed form, normal approximation):

- **The randomization test removes F1's null bias**, since the bias is in $\hat{K}$.
- **Its power is that of the randomized contrast, whatever statistic it is applied to**: 0.392, 0.391 and 0.393 for F1, VF and S in the Hybrid design at $f_m = 0.4$, against 0.627 for the model-based F1 test and 0.400 for VF; 0.44 against 0.88 for F1 in OL+BDC; 0.35 against 0.36 in CO, where the within-path information is already balanced by the crossover (Figure 1, panel D). The randomization test does not throw information away arbitrarily: the extra power of F1 is exactly the information whose validity depends on modeling the natural course and the expectancy curve.

The weak null (Proposition 7; derived). With heterogeneity the paths have covariances $\Sigma_p = \Sigma + \tau^2 \Psi_p \Psi_p'$, and the permutation distribution of $\hat{C}$ has the variance of the path mixture, $N^{-1} \sum_p \varpi_p (\omega_p - \bar{\omega})' \bar{\Sigma} (\omega_p - \bar{\omega})$ with $\bar{\Sigma} = \sum_r \varpi_r \Sigma_r$, while its sampling variance uses Σ_p on path p. The difference is

$$\frac{\tau^2}{N} \Big[\sum_p \varpi_p \{(\omega_p - \bar{\omega})' \Psi_p\}^2 - \sum_p \varpi_p \sum_r \varpi_r \{(\omega_p - \bar{\omega})' \Psi_r\}^2 \Big],$$

positive when the heaviest-weighted paths have the largest exposure, which makes the unstudentized test liberal. With two equally allocated paths, $\omega_2 - \bar{\omega} = -(\omega_1 - \bar{\omega})$ and the two sums are equal, so the unstudentized test is asymptotically exact, the multivariate form of the known result for the two-sample difference in means with equal group sizes [28] (derived). In the Hybrid design the variance ratio is 1.007 at $\tau = 2.75$ and 1.027 at $\tau = 5.49$, which raises the size of the unstudentized test to 0.051 and 0.053; in OL+BDC and CO it is 1 (verified, `58-tg-weak-null-permutation.csv`).

Studentization. A permutation test of a studentized statistic is asymptotically valid under the weak null when the studentizing variance is consistent for the statistic's sampling variance and the statistic is asymptotically linear: Janssen [30] for the Behrens-Fisher problem, Chung and Romano [28] in general, DiCiccio and Romano [31] for regression coefficients, and Wu and Ding [29] for Fisher randomization tests of weak nulls in completely randomized, stratified and clustered experiments, which they show are exact under the sharp null and asymptotically conservative under the weak null (quoted in substance from the full text read). Here the units are participants, the arms are paths, and the statistic is a linear contrast of arm means of vector outcomes, so the conditions are those of a multi-arm completely randomized experiment with a cluster-robust studentization (argued). Two cautions (derived):

- **Studentize the contrast, not the whole F1 estimate.** For F1, $t = (\hat{C} + \hat{K}) / \widehat{\mathrm{SE}}$ with $\hat{K}$ fixed and of order b_0; under the permutation distribution the standard error changes but $\hat{K}$ does not, so $\hat{K} / \widehat{\mathrm{SE}}$ varies with the relabeling and the studentized F1 statistic is not pivotal unless $b_0 = 0$. VF, whose $\hat{K}$ is zero, has no such problem; it is the natural statistic for a randomization test.
- **The weak null is not the sharp null with noise.** Under the weak null some participants are helped and some harmed. A randomization test that rejects says the mean effect is not zero only through the asymptotic argument; under the sharp null it says that someone responded, which is a weaker conclusion than paper 04 needs (argued; cf. Ding, Feller and Miratrix [32] on testing effect variation).

Selection inside the relabelings. Because exactness holds for any function of $(\mathbf{Y}, \mathbf{L})$, the AIC choice, the maximum over $\mathcal{T}$ and the averaging can be run inside every relabeling. The randomization p-value of F1A is then exact under the sharp null, which answers the face-validity request (the data choose the half-life) without the size cost of Proposition 5.

9.4 AIC-weight averaging and the estimand

Model averaging with smoothed AIC weights [36] over $\mathcal{T}$ has two problems for the main effect (derived):

- **The averaged quantity must be common to the models.** The coefficients of $D_{bc}(h)$ for different h are on different scales (Section 4.2): at true $t_{1/2} = 0.2$ the F1 limits are -2.74 at $h = 0$ and -3.62 at $h = 2$ while the model-implied exposure contrasts are -2.74 and -2.00 (Table 4). Averaging coefficients averages different estimands. The model-implied exposure contrast $\hat{\beta}_D^{\exp}(h) = \hat{\beta}_D(h)\bar{d}(h)$ is common to all h and targets $\beta_D^{\exp}$; it is the quantity to average, with Buckland's unconditional standard error $\sum_h w_h\{\widehat{\mathrm{SE}}_h^2 + (\hat{\beta}_D^{\exp}(h) - \bar{\beta})^2\}^{1/2}$.
- **The weights see the tested term entirely.** By Proposition 5, $w_h \propto \exp\{\ell_1(h)\} \propto \exp\{\mathrm{LR}(h)/2\}$: the weights are a smooth version of F1A's selection, with no null-model alternative. Averaging is therefore a smoothed max-type statistic referred to a single-t reference. Its size is between that of a fixed h and that of F1A, and it is not known in closed form; it is checked in Section 11. Hjort and Claeskens [37] show that ignoring the selection step gives intervals that are too narrow, and the same applies to averaging with data-dependent weights unless the weights are accounted for.

Averaging is a reasonable way to report $\beta_D^{\exp}$ across half-lives; it is not a test with known size.

9.5 Parametric bootstrap under the null

For the main effect the null model has no half-life, so the bootstrap is simpler than in `docs/57`, Section 4.5 (derived): fit $Y \sim t + \eta_t$ with random intercept and CAR(1) once, simulate $\mathbf{Y}_i^*$ from it for the observed participants, visits and paths, and rerun the whole procedure (fixed h, F1A, maximum, averaging) on each resample. It calibrates the selection exactly under the fitted null model, and it does not remove a null bias that the null model reproduces: the bootstrap data have the null model's linear-time and η_t mean, which F1 fits exactly, so F1's bootstrap null limit is zero while its real null limit is $b_0 = +0.33$ (derived). The bootstrap of F1 is therefore centered in the wrong place by the same amount as the model-based test; the bootstrap of VF is not affected. For a simulation study the warp-speed method of Giacomini, Politis and White [38], one resample per replicate, estimates the bootstrap test's rejection rate at about twice the cost of the plain procedure.

9.6 Joint estimation of the half-life

Identification only under the alternative (derived). In the partially linear model with mean $\beta_D\,\mathbf{x}_p(h) + Z\xi_Z$, the information about h is proportional to β_D^2 [42]: $I_{hh} = \beta_D^2 \sum_i \dot{\mathbf{x}}'_{p(i)} V^{-1} \dot{\mathbf{x}}_{p(i)}$ after the other columns are partialled out, with $\dot{x}_{pt} = \partial x_{pt}/\partial h = x_{pt} \ln 2\, t_{sd,pt}/h^2$ off drug. Two singularities follow:

- **At** $\beta_D = 0$ the information is zero (Section 9.1).
- **At** $h \to 0$ the derivative vanishes faster than any power of h, because the smallest t_{sd} is one week: every h below about 0.15 week gives the same regressor. The parameter $r_1 = 2^{-1/h}$ removes this singularity: $x_{pt} = r_1^{t_{sd,pt}}$, a polynomial with $\partial x_{pt}/\partial r_1 = 1$ at the first off-drug visit of each run and 0 elsewhere when $r_1 = 0$. In r_1 the boundary at 0 is regular, and by Self and Liang [39] the likelihood-ratio statistic for $r_1 = 0$ is asymptotically a 50:50 mixture of χ_0^2 and χ_1^2, so the one-sided profile interval uses the deviance cut 2.71 rather than 3.84 (derived; conditional on $\beta_D \neq 0$ and away from $|\beta_D|$ small).

How much a trial says about h. Table 9 gives the expected ML deviance of F1, $N \sum_p \varpi_p[\mathbf{r}_p(h)' V^{-1} \mathbf{r}_p(h)]$ minus its minimum over h, a first-order approximation to the expected profile deviance, and the best-fitting value h^* (verified, `58-cf-profile-deviance.csv`).

Table 9: Expected ML deviance of F1 against the analysis half-life, Hybrid design, $N = 70$, and the best-fitting half-life h^* on a fine grid (verified, first-order approximation)

True $t_{1/2}$	f_m	h^*	dev. at $h = 0$	0.2	0.5	1	2	∞
0.2	0.4	0	0	0.03	0.40	1.03	1.69	4.25
0.2	1	0	0	0.24	3.19	8.52	14.01	31.47
0.5	0.4	0.5	0.10	0.07	0	0.12	0.41	2.63
0.5	1	0.35	0.36	0.22	0.20	2.10	5.20	20.65
1	0.4	2.4	0.71	0.67	0.40	0.13	0.01	1.46
1	1	1.2	2.69	2.42	0.90	0.03	0.31	11.14
2	1	6	5.15	4.98	3.70	2.11	0.78	4.93

Readings (verified):

- **At the protocol's alternative ($f_m = 0.4$) a trial of 70 says almost nothing about** h: no finite h is more than 2 deviance units from the best when the truth is short, so every profile interval would cover 0 to 2 weeks.
- **At the published ceiling it separates short from long**: a truth of 0.2 week excludes $h \geq 1$ (deviance 8.5), as in the interaction example of `docs/57`, Section 5.
- **The best-fitting h is a working value that depends on the effect size**: for a true one-week half-life, $h^* = 2.4$ at $f_m = 0.4$ and 1.2 at $f_m = 1$; for a true 2 weeks, $h^* = 6$. Under F1 the null bias $b_0(h)$ and the Gompertz onset both pull the fit, and their relative weight changes with f_m (derived from Proposition 1). A data-chosen h must be reported as a working value, not as the drug's carryover half-life.

Testing β_D with h estimated. The profile likelihood of β_D maximizes over h, and at $\beta_D = 0$ the profile is flat in h, so the profile LR statistic for $\beta_D = 0$ is $\max_h \mathrm{LR}(h)$, the supremum statistic (derived from Proposition 5). Referring it to χ^2_1, as a profile-likelihood interval for β_D does, is the AIC problem again; the interval is too short near zero. Away from zero, with β_D well determined, h is identified and the usual profile intervals apply, with h still estimating h^*. The Wald test at the joint estimate has the same defect, and no CR2-type small-sample correction exists for the nonlinear parameter (`docs/57`, Section 4.6).

9.7 Confidence intervals that account for the half-life choice

Three constructions (derived):

1. **Simultaneous intervals over the grid** (in the spirit of PoSI, Berk et al. [35]). With $c_{\max}$ of Section 9.2, the intervals $\hat{\beta}_D^{\exp}(h) \pm c_{\max} \widehat{\mathrm{SE}}\{\hat{\beta}_D^{\exp}(h)\}$ cover their own targets $\operatorname{plim} \hat{\beta}_D^{\exp}(h)$ simultaneously with probability about $1 - \alpha$, so the interval at any data-chosen h, including the AIC choice, covers its target. Their union covers the target of whichever h is reported. They do not remove the gap between a target and $\beta_D^{\exp}$ (Table 4).
2. **Inversion of the randomization test** under constant-shape sharp hypotheses $H(b, h)$: everyone's drug effect is $b\, D_{bc,pt}(h)$. Under $H(b, h)$ the adjusted outcomes $\mathbf{Y}_i - b\, \mathbf{x}_{p(i)}(h)$ are exchangeable across labels, so the set of b not rejected is an exact interval for b given h, and its union over h, converted with $\bar{d}(h)$, is an interval for the exposure contrast that is valid if the true effect has one of the grid's shapes. The true Gompertz onset is not of that shape, so the coverage under the refined process is approximate (argued).
3. **Maximized p-values** (Berger and Boos [33]; Ding, Feller and Miratrix [32]): for a hypothesized $b \neq 0$, h is identified, so the p-value can be maximized over a confidence set for h of level $1 - \varsigma$ and ς added. This is valid for inverting tests of $b \neq 0$; at $b = 0$ it reduces to the randomization test, where h plays no role.

10. Worked example: one trial with a drug effect

10.1 Set-up

One Hybrid trial was drawn from the refined process of the stress-test driver (anchored carryover, randomized allocation, separable AR(1) at $\rho = 0.7$), $N = 70$, path sizes 14, 18, 18 and 20, with a drug effect $f_m = 0.4$ ($m_{BR} = 4.39$ points), true $t_{1/2} = 0.2$ week and no heterogeneity. The estimands are $\beta_D^{\text{exp}} = -2.85$ and $\beta_D^{\text{bl}} = -2.99$ points. The seed (5801) was fixed before the trial was generated, and the trial was not selected for any property (inspected, `58-randomization-example.R`).

10.2 The statistic across the half-life

Table 10: Example: F1 and VF at each grid half-life, REML with CR2 and Satterthwaite degrees of freedom; the model-implied exposure contrast is $\hat{\beta}_D(h)\bar{d}(h)$ (verified, `58-ex-grid.csv`; VF degrees of freedom 66.4 to 67.5)

h (weeks)	F1 $\hat{\beta}_D$	CR2 SE	$T(h)$	df	p	$\bar{d}(h)$	F1 $\hat{\beta}_D^{\text{exp}}$	VF $\hat{\beta}_D$	VF $T(h)$	VF p
0	−2.604	1.181	−2.204	68.9	0.031	1	−2.604	−2.883	−1.635	0.107
0.2	−2.637	1.198	−2.201	68.9	0.031	0.991	−2.612	−2.942	−1.631	0.108
0.5	−2.771	1.326	−2.090	68.8	0.040	0.907	−2.514	−3.146	−1.504	0.137
1	−2.825	1.538	−1.837	68.9	0.070	0.757	−2.138	−2.974	−1.219	0.227
2	−3.084	1.890	−1.631	68.9	0.107	0.554	−1.710	−2.894	−0.971	0.335

Readings (verified):

- **The coefficient grows with h while the exposure contrast shrinks**: from −2.60 to −3.08 against −2.60 to −1.71, the scale effect of Section 4.2. At $h \leq 0.5$ the exposure contrast is within 0.34 point of the estimand.
- **The profile cannot locate h**: the ML estimate is on the boundary (0.020 week on a continuous search), the deviance never exceeds 2.1 at any h (1.40 at 1 week, 1.29 as $h \to \infty$), and the profile interval is all of $[0, \infty)$ at either cut (Figure 2B), as Table 9 predicts for $f_m = 0.4$.
- **The likelihood ratio for the drug term is largest at $h = 0$** (4.27), so F1A chooses $h = 0$; in this trial that changes nothing, because $T(0)$ and $T(0.2)$ differ by 0.003.

10.3 What each procedure reports

Table 11: Example: what each procedure reports for the same trial (verified, `58-ex-*.csv`; the null-model-weight statistics use CR0 studentization, hence $|t| = 2.13$ rather than 2.20)

Procedure	h used	$\hat{\beta}_D^{\text{exp}}$	Statistic	p
F1, fixed $h = 0.2$, CR2	0.2	−2.61	$t = -2.20$	0.031
F1A, AIC with the tested term (invalid reference)	0	−2.60	$t = -2.20$	0.031
F1, AIC-weight average on the β_D^{exp} scale	weighted mean 0.50	−2.43	$t = -2.01$	0.048

Procedure	h used	$\widehat{\beta}_D^{\exp}$	Statistic	p
F1, max-$\lvert t\rvert$, Bonferroni	0		2.20	0.154
F1, max-$\lvert t\rvert$, multivariate normal ($c_{\max} = 2.19$)	0		2.20	0.047
F1, max-$\lvert t\rvert$, Davies' bound on $[0, 4]$			2.20	0.049
Randomization, F1 at $h = 0.2$, null-model weights (999 permutations; 999 re-randomizations)	0.2		$\lvert t\rvert = 2.13$	0.048; 0.050
Randomization, F1 max-$\lvert t\rvert$ with selection inside (same)			2.14	0.056; 0.064
Randomization, F1 at $h = 0.2$ and F1A, with refits (49 permutations)	0.2; 0		2.20	0.08; 0.08
Parametric bootstrap from the null model, F1 at 0.2; F1A (49 resamples)			2.20	0.08; 0.10
VF, fixed $h = 0.2$, CR2	0.2	-2.91	$t = -1.63$	0.108
VF, max-$\lvert t\rvert$, multivariate normal ($c_{\max} = 2.15$); Davies	0		1.64	0.151; 0.186
Randomization, VF at $h = 0.2$ (999; 999)	0.2		1.53	0.121; 0.113
Randomization, VF max-$\lvert t\rvert$ with selection inside (999; 999)			1.53	0.182; 0.160

Table 12: Example: 95% intervals for the exposure contrast that do and do not account for the half-life choice (verified, `58-ex-intervals.csv`; inversion on a 0.25-point grid of b)

Interval for $\beta_D^{\exp}$ (truth -2.85)	Lower	Upper
F1, fixed $h = 0.2$, CR2	-4.98	-0.24
F1, AIC choice, naive	-4.96	-0.25
F1, AIC choice with the max-$\lvert t\rvert$ constant	-5.19	-0.02

Interval for $\beta_D^{\exp}$ (truth -2.85)	Lower	Upper
F1, AIC-weight average (Buckland)	-4.84	-0.02
F1, union over $\mathcal{T}$ with the max-$\lvert t\rvert$ constant	-5.21	0.58
VF, fixed $h = 0.2$, CR2	-6.48	0.65
VF, union over $\mathcal{T}$ with the max-$\lvert t\rvert$ constant	-6.94	1.95
VF, randomization inversion at $h = 0.2$ (199 permutations)	-6.69	0.74
VF, randomization inversion, union over $\mathcal{T}$	-6.80	1.80

Readings (verified unless stated):

- **The verdict depends on what the analysis is allowed to assume**, not on the half-life. Every F1 procedure that relies on the time and expectancy model rejects at or near 0.05 (0.031 at a fixed h, 0.047 to 0.056 once the choice of h is paid for); every procedure that relies only on the randomized between-path contrast (VF, its calibrated maximum, the VF randomization tests) does not (0.108 to 0.186). The half-life choice moves p by 0.02 at most within either family.
- **The randomization test of the F1 statistic agrees with the calibrated F1 tests** (0.048 at a fixed h, 0.056 with the maximum inside), because its direction comes from the invariant within-path part (Section 11.3). The refit version with 49 permutations gives 0.08, which with 49 relabelings has a resolution of 0.02 and a Monte Carlo error of about 0.04 (derived); it was cut from 199 because the machine was heavily loaded (Section 10.4), and it is not evidence against the cheap version.
- **The bootstrap of F1 is centered in the wrong place by design** (Section 9.5); with 49 resamples its p-values (0.08, 0.10) are too imprecise to compare.
- **Accounting for the choice widens the intervals modestly**: the F1 interval at the AIC choice goes from $[-4.96, -0.25]$ to $[-5.19, -0.02]$ with the max-$\lvert t\rvert$ constant; the union over the grid crosses zero. All intervals cover the truth in this trial, which is one trial.

10.4 Run time

Grids of F1 and VF at five half-lives: 1.7 seconds; the fine profile (54 half-lives, F1 and VF, REML and ML): 17 seconds; the label-invariant randomization tests (999 permutations and 999 re-randomizations of ten statistics): 1.9 seconds; the refit randomization test (49 permutations): 324 seconds and the null-model bootstrap (49): 25 seconds, the same work, so the difference measures the machine load at the time; the randomization inversion (265 hypotheses, 199 permutations each): 8 seconds; 379 seconds in all on 7 workers (verified, `58-ex-timing.csv`). A first run with 199 refits and resamples was stopped after 15 minutes under a load average near 70.

11. A modest Monte Carlo check

11.1 Set-up

Four Hybrid cells of the refined process at $N = 70$ (anchored carryover, randomized allocation, separable AR(1) at $\rho = 0.7$): the sharp null ($f_m = 0$, $\tau = 0$, $t_{1/2} = 0.2$), a weak null ($f_m = 0$, $\tau = 5.49$), and alternatives at $f_m = 0.4$ with true $t_{1/2} = 0.2$ and 1 week. Per trial: F1 and VF at every $h \in \mathcal{T}$ (REML, CR2), the AIC choice with the tested term, AIC averaging on the $\beta_D^{\exp}$ scale, the maximum over the grid with Bonferroni and with the influence-based multivariate normal calibration, and the cheap exact randomization

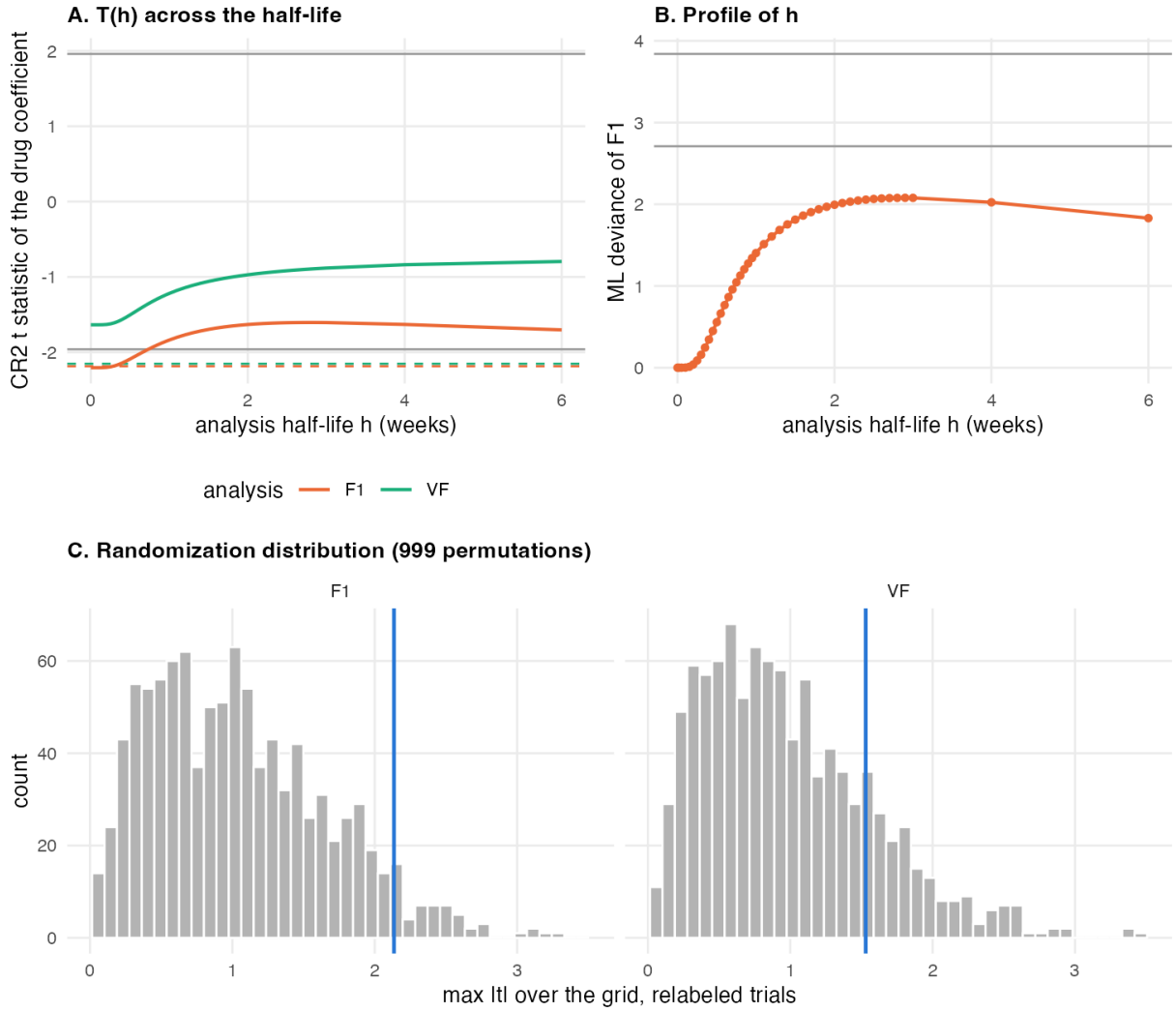


Figure 2: Example trial ($N = 70$, $f_m = 0.4$, true $t_{1/2} = 0.2$ week, refined process). A: CR2 t-statistic of the drug coefficient for F1 and VF over a fine grid of analysis half-lives, with ± 1.96 (grey) and $-c_{\max}$ for each analysis (dashed). B: ML deviance of F1 against h, with the cuts 2.71 and 3.84. C: the randomization distribution of $\max_h |T(h)|$ over 999 permutations of the path labels with label-invariant weights, with the observed value (vertical line).

tests of Section 9.3 with 199 relabelings at fixed path counts (inspected, `58-mc-check.R`). Seeds are per trial, $58{,}000{,}000 + 100{,}000 \times \text{cell} + r$.

Replicates. The run was planned at 1,000 null and 500 alternative trials per cell and was cut to 600 and 300 while running, to respect the time budget; 1,800 trials were analyzed (600, 600, 300, 300). The MCSE of a rate near 0.05 is 0.0089 per null cell, and of a power near 0.5 at most 0.029. Differences from the closed form smaller than about 0.02 in size, or 0.06 in power, are not resolved. This is a check of the closed form, not a size study.

11.2 Results

Table 13: Monte Carlo check, Hybrid, $N = 70$: moments of the drug coefficient and rejection rates at 0.05, with the closed-form value in parentheses where Sections 5 to 9 give one (verified, `58-mc-summary-*.csv`; MCSE 0.0089 for null rates near 0.05, at most 0.029 for power)

Quantity	Sharp null	Weak null	Alt., $t_{1/2} = 0.2$	Alt., $t_{1/2} = 1$
n_{sim}	600	600	300	300
F1, $h = 0.2$: mean estimate (closed-form limit)	0.329 (0.327)	0.388 (0.327)	−2.907 (−2.783)	−1.652 (−1.629)
F1, $h = 0.2$: empirical SD (closed-form SE)	1.245 (1.219)	1.329 (1.304)	1.204 (1.219)	1.283 (1.216)
F1, $h = 0.2$: mean CR2 SE; mean CR2 df	1.216; 68.9	1.303; 68.9	1.216; 68.9	1.216; 68.9
VF, $h = 0.2$: mean estimate (limit)	0.011 (0)	0.075 (0)	−2.907 (−2.830)	−1.587 (−1.680)
VF, $h = 0.2$: empirical SD (closed-form SE)	1.714 (1.661)	1.759 (1.714)	1.701 (1.661)	1.752 (1.656)
F1, fixed $h = 0.2$	0.065 (0.058)	0.060 (0.057)	0.653 (0.627)	0.263 (0.269)
F1A, AIC with the tested term	0.093 (0.088)	0.085	0.707 (0.666)	0.400 (0.407)
F1, AIC-weight average	0.053	0.062	0.612	0.317
F1, max-$\lvert t \rvert$, Bonferroni	0.020	0.027	0.428	0.187
F1, max-$\lvert t \rvert$, multivariate normal	0.060 (0.056)	0.072	0.639 (0.590)	0.340 (0.331)
VF, fixed $h = 0.2$	0.058 (0.050)	0.055 (0.050)	0.400 (0.400)	0.167 (0.172)
VFA, AIC with the tested term	0.087 (0.077)	0.083	0.420 (0.423)	0.197 (0.218)
VF, AIC-weight average	0.050	0.062	0.363	0.163
VF, max-$\lvert t \rvert$, multivariate normal	0.058 (0.051)	0.068	0.383 (0.351)	0.183 (0.169)
Randomization, F1 $\lvert t \rvert$ at $h = 0.2$	0.063	0.053	0.507	0.200

Quantity	Sharp null	Weak null	Alt., $t_{1/2} = 0.2$	Alt., $t_{1/2} = 1$
Randomization, F1 $\lvert\widehat{\beta}\rvert$ at $h = 0.2$ (unstudentized)	0.067	0.067	0.507	0.203
Randomization, F1 max-$\lvert t\rvert$ (selection inside)	0.065	0.062	0.417	0.180
Randomization, VF $\lvert t\rvert$ at $h = 0.2$	0.065	0.052	0.393 (0.391)	0.163
Randomization, VF $\lvert\widehat{\beta}\rvert$ (unstudentized)	0.060	0.067 (0.053)	0.420	0.173
Randomization, VF max-$\lvert t\rvert$ (selection inside)	0.055	0.060	0.370	0.167

Table 14: Share of trials in which AIC (with the tested term) chose each analysis half-life (verified, `58-mc-aic-choice.csv`)

Cell	Analysis	$h = 0$	0.2	0.5	1	2
sharp null	F1A	0.343	0.055	0.112	0.120	0.370
sharp null	VFA	0.363	0.038	0.108	0.105	0.385
alt., $t_{1/2} = 0.2$	F1A	0.567	0.113	0.077	0.057	0.187
alt., $t_{1/2} = 1$	F1A	0.237	0.027	0.097	0.120	0.520

11.3 Reading

- **The closed form holds** (verified). F1's mean estimate under the sharp null is 0.329 against the limit 0.327, and its empirical SD 1.245 against 1.219 (2% larger); VF's mean is 0.011 against 0, and its SD 3% above the closed form. Power agrees within one to two MCSE in every cell checked (0.653 against 0.627, 0.400 against 0.400, 0.263 against 0.269, 0.167 against 0.172). The mean CR2 standard errors match the closed-form SEs to within 0.3%, and the CR2 Satterthwaite degrees of freedom average 69 for F1 and 66 to 67 for VF, so the normal reference is adequate (Section 8).
- **AIC selection with the tested term is liberal, as Proposition 5 says**: 0.093 and 0.087 under the sharp null (closed form 0.088 and 0.077), 0.085 and 0.083 under the weak null. Its choices pile up at the ends of the grid, 0 and 2 weeks (71% to 75% of null trials), the signature of a maximum over correlated statistics whose ends are the least correlated. Under the alternative with $t_{1/2} = 1$ week it chose 2 weeks in 52% of trials, which is h^* territory (Table 9).
- **F1 at a fixed h is close to, and slightly above, nominal**: 0.065 (closed form 0.058) under the sharp null and 0.060 under the weak null, consistent with the null bias of Section 5.2 within MC error. VF at a fixed h is at 0.058 and 0.055.
- **The calibrated maximum and averaging are near nominal; Bonferroni is conservative**: max-$\lvert t\rvert$ with the multivariate normal 0.060 (F1) and 0.058 (VF) under the sharp null, 0.072 and 0.068 under the weak null; averaging 0.053 and 0.050, 0.062 under the weak null; Bonferroni 0.020. The calibrated maximum costs 0.014 to 0.017 of power relative to the fixed h when the fixed h is right ($t_{1/2} = 0.2$) and gains 0.016 to 0.077 when it is wrong ($t_{1/2} = 1$).
- **The randomization tests are exact by construction, and the rejection rates are consistent with that within MC error**: 0.055 to 0.067 under the sharp null for the eight randomization statistics, 1 to 2 MCSE above 0.05, on the same trials on which the model-based VF test rejected 0.058; with 199 relabelings the exact size is 0.050 (derived), so these rates indicate a chance excess shared by the trials, not a defect, but they cannot show exactness more sharply than 0.018.

- **The randomization power matches Proposition 6** for VF (0.393 against 0.391). For the F1 statistic it is higher than predicted, 0.507: because $\widehat{K}$ is invariant and carries the within-path estimate, $|\widehat{C} + \widehat{K}|$ behaves as a one-sided test of $\widehat{C}$ in the direction of $\widehat{K}$, at 0.05 one-sided rather than 0.025 per side; a one-sided test with $\Lambda \approx 1.68$ has power 0.51 (derived). This is legitimate, since the direction is chosen by a label-invariant quantity, and it is the one way the within-path information can enter an exact test.
- **The weak null** (single cell, $\tau = 5.49$): the unstudentized VF randomization test rejected 0.067 against a predicted 0.053, the studentized one 0.052; the difference is within 2 MCSE and does not resolve Proposition 7.

11.4 Run time

The 1,800 trials took 44,050 worker-seconds as recorded in the chunks, but that figure is not a property of the procedure: the first launch ran with a multithreaded BLAS, which oversubscribed the cores, and the machine was heavily loaded by other processes (load average 50 to 70) and appears to have been suspended for part of the run. A single trial with one BLAS thread on an unloaded worker takes 1.7 seconds for the whole per-trial procedure (verified), so the check would take about $1{,}800 \times 1.7/7 \approx 7$ minutes on 7 workers (derived). The elapsed time from the first launch to the last chunk was about two hours, well over the 45-minute budget; the replicate counts were cut as stated above and no chunk was rerun.

12. Comparison and provisional recommendation

12.1 Comparison table

"Sharp" and "weak" give validity under the two nulls of Section 4.3. Size and power are for the Hybrid design at $N = 70$, sharp null and base alternative ($f_m = 0.4$, $t_{1/2} = 0.2$): closed form (cf, normal approximation) and the Monte Carlo check (MC, Section 11; MCSE 0.009 for size, 0.028 for power). "Face validity" asks whether the data choose or check the half-life.

Table 15: Methods for testing the drug main effect with an unknown carryover half-life, Hybrid design, $N = 70$ (cf: closed form, verified; MC: Section 11, verified; validity entries derived or argued as in Sections 5 to 9)

Method	Sharp null	Weak null	Size, cf / MC	Power, cf / MC	Cost per trial	Estimand	Face validity	Status
F1, h fixed at 0.2	approximate null bias +0.33, size grows with N	as sharp; CR2 includes τ^2	0.058 / 0.065	0.627 / 0.653	1 fit	β_D^{exp} via $\bar{d}(h)$, 3% off	weak	cf, MC
VF, h fixed at 0.2	null limit exactly 0	as sharp	0.050 / 0.058	0.400 / 0.400	1 fit	randomized contrast	weak	cf, MC
F1A, AIC with the tested term	invalid (Proposition 5)	invalid	0.088 / 0.093	0.666 / 0.707, inflated	10 fits	at the chosen h	good	not recommended

Method	Sharp null	Weak null	Size, cf / MC	Power, cf / MC	Cost per trial	Estimand	Face validity	Status
Max-$\|t\|$ over $\mathcal{T}$, F1, multivariate normal	approximate, keeps F1's bias	approximate	0.056 / 0.060	0.590 / 0.639	5 fits	none single	moderate	cf, MC
Max-$\|t\|$ over $\mathcal{T}$, VF, multivariate normal	approximate	approximate	0.051 / 0.058	0.351 / 0.383	5 fits	none single	moderate	cf, MC
Max-$\|t\|$, Bonferroni	conservative	conservative	below 0.02 / 0.020	not computed / 0.428	5 fits	none single	moderate	MC
Max-$\|t\|$, Davies' bound, continuous h	approximate, nearly exact here	approximate	bound tight to 0.003 in $c_{\max}$	as multivariate normal	fine grid of fits	none single	moderate	cf, example
Randomization, VF statistic, maximum over $\mathcal{T}$ inside	exact	asymptotic if studentized	0.050 by construction / 0.055	0.39 / 0.370	1 null fit; 2 s for 999 relabelings	randomized contrast	good	cf, MC, example
Randomization, F1 statistic	exact	studentized form not pivotal (Section 9.3)	0.050 by construction / 0.063	0.39 two-sided theory / 0.507	as above	F1's, direction from within-path part	good	MC, example
AIC-weight averaging on the $\beta_D^{\exp}$ scale	unknown; weights are $\exp\{\mathrm{LR}/2\}$	unknown	not derived / 0.053	not derived / 0.612	10 fits	average of $\widehat{\beta}_D^{\exp}(h)$	good	MC
Parametric bootstrap from the null model	calibrates the selection; not F1's bias	approximate	not run at scale	not run	M times 10 fits	as the statistic	good	example ($M = 49$)
Joint nonlinear estimation of h	invalid near $\beta_D = 0$ (supremum)	as sharp	not run	not run	1 `nlme` fit	h^*, a working value	good	proposed, not recommended as a test

Method	Sharp null	Weak null	Size, cf / MC	Power, cf / MC	Cost per trial	Estimand	Face validity	Status
Sample splitting (choose h on one half)	exact for the selection	as fixed h	not run	about a $\sqrt{2}$ loss of noncentrality (`docs/57`)	2 half fits	as fixed h	good	proposed
Bayesian marginalization over h	unknown; needs calibration	unknown	not run	not run	MCMC	posterior of β_D	good	proposed
Intervals: max-$\lvert t\rvert$ constant (PoSI-type)	approximate simultaneous coverage	approximate			5 fits	target at each h	good	example
Intervals: randomization inversion, union over $\mathcal{T}$	exact if the effect has a grid shape				265 tests of 199 relabelings, 8 s	exposure contrast	good	example

12.2 Provisional recommendation for paper 04

Provisional; it rests on closed-form results, 1,800 Monte Carlo trials in four Hybrid cells and one example trial.

1. **Primary test: F1 with $t^a_{1/2} = 0.2$ week fixed in advance**, CR2 with Satterthwaite degrees of freedom, as the protocol proposes. A fixed h in the pharmacological range costs almost nothing when the truth is in that range (Table 4), and no data-chosen rule can gain much power where the data barely identify h (Table 9). Its null bias (+0.33 points in the Hybrid design) must be reported and its size checked at $N = 150$, where the closed form predicts 0.068; if the stress test confirms an excess there, VF becomes the primary analysis.
2. **Size-exact companion with data-chosen half-life: a randomization test of the VF statistic with the maximum over $\mathcal{T}$ computed inside each relabeling**, fixed-count permutation, 999 relabelings, label-invariant null-model weights, CR0 studentization. It answers the face-validity request (the data choose h, and the reader sees at which h the evidence is strongest) with exact size under the sharp null, and it states plainly how much of the evidence rests on the randomization alone.
3. **Report, in every trial**: $\hat{\beta}^{\exp}_D(h)$ across $\mathcal{T}$ with CR2 intervals; the AIC-weighted average of these as a descriptive summary, not a test; the profile deviance of h in the r_1 parameterization with the 2.71 cut; and the statement that the best-fitting h is a working value (Section 9.6).
4. **Not to be used as tests**: F1A or VFA with a nominal reference; S (linear time without η_t) in the Hybrid or OL+BDC design; the bootstrap of F1 as a correction for its null bias; joint estimation with a χ^2_1 reference near zero.

12.3 Proposed edits to the protocol (not applied)

The protocol was not edited. Proposed edits, for the user's decision:

- **Section 4.2, estimand.** Add that every $D_{bc}(h)$ analysis reports the model-implied exposure contrast $\hat{\beta}_D(h)\bar{d}(h)$ against β_D^{exp}, and that bias and coverage are computed on that scale, because the coefficient's scale changes with h (Section 4.2).
- **Section 5.1, closed form.** Use the ML limit of the random-intercept CAR(1) working covariance rather than working independence, include the null bias $b_0(h)$ and the mean-misfit term of the sandwich, and replace "a nonzero limit is predicted for linear time without η_t" by the Table 3 values, which show a nonzero limit for F1 as well.
- **Section 5.2, analysis list. Add:** RV, the randomization test of VF with the maximum over $\mathcal{T}$ inside (999 fixed-count permutations, null-model weights, all cells); RF, the same with the F1 statistic; MX-F1 and MX-VF, the max-$|t|$ tests over $\mathcal{T}$ calibrated by the influence-based multivariate normal; F1A as a labeled negative control (its size is predicted at 0.088). **Amend:** the "Misspecified half-life" note to use $\mathcal{T} = \{0, 0.2, 0.5, 1, 2\}$ in every cell (the cost is five fits).
- **Section 6.1, measures.** Add the h at which $|T|$ is largest and the AIC choice, by cell; the profile-interval width in r_1.
- **Section 6.4, decision rule.** Add a size screen at $N = 150$ for the analyses with a nonzero closed-form null bias (F1, S), since a fixed bias inflates size with N.
- **Section 8, open decisions. Revise item 12 (L1)** to the label-invariant randomization test above, run in every cell rather than a subset, since it costs about 2 seconds per trial. **Add:**
  (21) whether paper 04's primary test is model-based (F1, power 0.63 at the base alternative) or randomization-based (RV, power about 0.39), or both reported; (22) the half-life policy: fixed at 0.2 week, or chosen inside a calibrated test (RV, MX); (23) whether the weak-null cells (block 2) judge the randomization tests in their studentized form only; (24) whether any main-effect result is quoted from runs with the compounding carryover recursion (Section 7 recommends not, at $t_{1/2} \geq 1$ week).

13. What the literature says

The protocol's Appendix A reviews 38 simulation and closed-form studies of main-effect tests in N-of-1, aggregated N-of-1, multi-crossover and single-case designs; its Section A.3 tables compare their data-generating processes and analyses. That review is not repeated. This section adds the literature on the four issues of this note and says how the development relates to it. The reading extent of each source is in Section 16.

Carryover in crossover trials. Grizzle [7] showed that in the two-period crossover the direct and residual (carryover) effects are estimable only with between-participant information, which led to the two-stage procedure: test carryover first, then analyze both periods or period 1 only. Freeman [8] showed that the preliminary test is highly correlated with the period-1 analysis, so the actual significance level exceeds the nominal level even without carryover and "increasing sample size only exacerbates the problem" (quoted, abstract); Senn [10, 12] and Fleiss [9] concluded that the procedure should not be used and that crossover analysis rests on assumptions about carryover, which washout must make plausible; Kabaila and Vicendese [13] found the same failure for a four-period design. The two-stage procedure is a data-chosen carryover model followed by a test that ignores the choice, the main-effect analog of F1A, and Proposition 5 is its mixed-model form: a selection over carryover regressors whose criterion is the tested term's own likelihood ratio. The textbooks of Senn [5] and Jones and Kenward [6] treat carryover as a nuisance that design should remove; Kenward and Jones [11] show that baseline-adjusted analyses need no assumption about the covariance structure, the stance CR2 takes here. The protocol's Appendix A.2.6 and A.2.10 (Cruz et al.; Sturdevant and Lumley) cover carryover estimation in repeated-measures crossovers.

Pharmacodynamic and lag models. The exponential decay of D_{bc} is the washout of a first-order effect compartment, the device of Sheiner et al. [14] (rate constant fitted from concentration and effect data), whose clinical use Holford and Sheiner [15] describe (record only). Percha et al. [17] simulate first-order wash-in and wash-out in single-person trials but do not model carryover in the analysis; Liao et al. [16] estimate a Bayesian distributed lag of fixed length chosen from the drug half-life; Gasparrini, Armstrong and Kenward [18] choose lag bases by an information criterion with no post-selection inference. Across this work the lag shape is either fixed in advance or chosen and then treated as known; none calibrates a test of the main

effect for the choice. The working-parameter result of Section 9.6 adds that, with Gompertz onset and a misspecified non-drug mean, the best-fitting decay is not the pharmacological one.

Nuisance parameters present only under the alternative. Davies [19, 20] formulated the problem and the upcrossing bound used in Section 9.2; Andrews and Ploberger [21] derived weighted-average-power optimal tests (average and exponential statistics); Hansen [22] gave a simulation method for the p-value of the supremum based on the conditional distribution of the score process, which is the multiplier version of the influence-based correlation used here. Seber and Wild [42] treat partially linear nonlinear models, in which the nonlinear parameter is identified only when its linear coefficient is not zero (record only). The boundary at $r_1 = 0$ is a case of Self and Liang [39].

Randomization tests. Exactness under the sharp null for any statistic is the classical randomization-model argument (Edgington and Onghena [23]; Ernst [40]); Rosenberger, Uschner and Wang [41] stress that the reference set must follow the actual, possibly restricted, randomization rule. In single-case research, randomization tests are designed around the randomization scheme (Onghena and Edgington [24]; Kratochwill and Levin [25]); Michiels et al. [26] wrap a multilevel model in a randomization test and keep the nominal level where the Kenward-Roger F test fails, and Bouwmeester and Jongerling [27] quantify the power cost of gradual onset. For the weak null, Janssen [30], Chung and Romano [28], DiCiccio and Romano [31] and Wu and Ding [29] show that permutation tests of unstudentized statistics need not control the level, while studentized versions are asymptotically valid (and exact under the sharp null); Ding, Feller and Miratrix [32] test treatment-effect variation with the mean effect as a nuisance, using the maximized p-value of Berger and Boos [33]. What this note adds is specific to the Hybrid design: the label-invariant null-model weights that make a mixed-model randomization test cheap (Section 9.3), the decomposition showing that such a test uses only the between-path contrast whatever statistic is permuted (Proposition 6), and the variance-ratio formula for the weak null with vector outcomes (Proposition 7).

Post-selection inference and model averaging. Leeb and Pötscher [34] show that inference ignoring selection is invalid and that the distribution of post-selection estimators cannot be estimated uniformly; Berk et al. [35] obtain valid intervals after any selection from a simultaneous max-$|t|$ constant, the construction of Section 9.7, item 1; Buckland, Burnham and Augustin [36] proposed AIC weights and an unconditional variance, and Hjort and Claeskens [37] show that naive intervals after selection or averaging are too narrow. For the main effect the selection is over a single nuisance index, so the PoSI constant is the $c_{\max}$ of five correlated statistics, near 2.16 rather than the Scheffé bound.

Small-sample inference and computation. CR2 with Satterthwaite degrees of freedom is from Bell and McCaffrey [3] and Pustejovsky and Tipton [4], who report size "nearly always less than or equal to the nominal" when the degrees of freedom exceed 4 or 5 (quoted); the warp-speed method is from Giacomini, Politis and White [38]. The variance-component results for series of N-of-1 trials (Senn [43]; Araujo, Julious and Senn [44]) give the sharp and weak null distinction its practical meaning: an analysis that ignores τ^2 is valid for the first and liberal for the second (protocol, Appendix A.4).

14. Limitations

- **First-order and normal approximations.** The probability limits are exact for the stated working covariance; the variances, sizes and powers of Sections 5 to 9 are normal approximations at expected path shares. The Monte Carlo check (Section 11) covers four Hybrid cells at $N = 70$ only.
- **One data-generating process.** The refined process with $c_{bm} = 0$, exponential decay, no dropout and no measurement error. The null bias of F1 depends on the shapes of the natural course and the placebo curve (Section 6), which the refined process fixes at the published Gompertz parameters; another shape would give another b_0.
- **Heterogeneity is the protocol's proposal** (v_i scaling the unit profile, independent of everything else), an open decision of the protocol (Section 8, item 9). The weak-null results depend on it.
- **The randomization tests use null-model weights.** The refit version was run on one trial only (Section 10). Their power under the weak null and with dropout is not evaluated; with dropout the cheap version needs per-pattern weights, and the refit version is the safe choice.

- **Not implemented:** joint nonlinear estimation of h with a sandwich, a Bayesian analysis that marginalizes over h, sample splitting for the main effect, the Andrews-Ploberger average statistics, and the coverage of any interval in Section 9.7 beyond the single example.
- **The recommendation is provisional** and depends on the paper's estimand decision (protocol, Section 8, item 1): a randomization test answers a question about the randomized contrast, and paper 04 may prefer to report the model-based F1 test with its known small bias.

15. How to reproduce

From the repository root, in this order (the figure script reads the outputs of the others):

```
Rscript analysis/scripts/main-effect-half-life/58-plim-and-power.R
Rscript analysis/scripts/main-effect-half-life/58-tgrid-correlation.R
export VECLIB_MAXIMUM_THREADS=1 OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1
Rscript analysis/scripts/main-effect-half-life/58-mc-check.R --cores 7 --reps-null 600 --reps-alt 300
Rscript analysis/scripts/main-effect-half-life/58-randomization-example.R --cores 7
Rscript analysis/scripts/main-effect-half-life/58-figures.R
```

All source `58-common.R`, which evaluates the top-level definitions of `analysis/scripts/quick-sim/carryover-closed-form` (everything before its pre-run checks, skipping the lines that create output directories), as `57-common.R` does, so the data-generating process is the driver's own, with anchored carryover and randomized allocation. The drug effect is scaled by f_m after the driver's `build()`, which is exact because the carryover rules are linear in the drug-response mean. Outputs go to `analysis/data/quick-sim/main-effect-half-life/`; figures to `docs/figures/58-*.png`.

- `58-common.R`: path moments, design matrices, the GLS limit and sandwich, the ML limit of the working covariance, estimands, the trial generator with heterogeneity, the main-effect fits with CR2, the influence-based correlation, the procedures, and the label-invariant randomization tests.
- `58-plim-and-power.R`: Tables 2 to 7 and 9, Figure 1 data (`58-cf-*.csv`).
- `58-tgrid-correlation.R`: Section 9.2 tables, Davies' bound, the weak-null variance ratio and the max-$|t|$ power (`58-tg-*.csv`).
- `58-mc-check.R`: Section 11 (`58-mc-*.csv`, chunks in `58-mc-chunks/`).
- `58-randomization-example.R`: Section 10 (`58-ex-*.csv`).
- `58-figures.R`: Figures 1 and 2.

Run times (verified, 4 performance plus 4 efficiency cores, 7 workers). 58-plim-and-power.R 16 seconds; 58-tgrid-correlation.R 15 seconds; 58-mc-check.R 1.7 seconds per trial unloaded, 1,800 trials, but about two hours elapsed under the conditions of Section 11.4; 58-randomization-example.R 379 seconds with 49 refits and resamples (Section 10.4); 58-figures.R a few seconds. The time budget of about 45 minutes was exceeded by the Monte Carlo check; its replicate counts were cut from 1,000 and 500 to 600 and 300, and the example's refit and bootstrap counts from 199 to 49.

16. References

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How the references were checked (2026-10-09). Bibliographic details of 2 to 42 were checked against at least one of PubMed, the Crossref record of the DOI (queried without any contact parameter), the publisher page or the Zotero item; 1, 43, 44 and 45 are as verified in the protocol and `docs/57`. Reading extent: full text, at least in part, for 3, 4, 16, 17, 18, 27, 28, 29, 32, 35, 38, 40 and 41 (16, 17, 18 and 27 through a summarizing reader of the PMC page, so their quotations may be paraphrased); abstract only for 7 to 14, 19 to 22, 24 to 26, 31, 33, 34, 36, 37 and 39; record only for 5, 6, 15, 23, 30 and 42. The Davies bound of Section 9.2 was checked against its implementation in `segmented::davies.test` (inspected), not against Davies' text. Statements attributed to a source read only in abstract are limited to what the abstract says.

Appendix A. Notation check

Every symbol used in this note, with its source. "Canonical" means `analysis/report/NOTATION.md`, Part 1; "protocol" means the working protocol's notation index; "defined here" means local to this note.

Table 16: Every symbol used, with its source

Symbol	Source
i, t, N, P, Y_{it}, BL_i, TV, PB, BR	canonical
η_t	canonical (design expectancy)
D_{it}, $D_{bc,it}$, t_{od}, t_{sd}, t_{pb}, $t_{1/2}$	canonical
$t^a_{1/2}$	canonical as used in `docs/56`, `docs/57` and the protocol
β_D, ρ, α, π, σ_{BR}	canonical
w_t, c_1	canonical
m_{BR}, τ^2	canonical (Gompertz ceiling of BR; heterogeneity variance)
Ψ_{it}, Ψ_p	canonical (2026-10-09; unit drug-response profile, formerly q_{it}, $\mathbf{q}_p$)
f_m, v_i, ϖ_p, p (path)	protocol; the path letter p follows the protocol and `NOTATION.md`'s q, p row, and coincides in letter with the p-value, which the text distinguishes by context
β_D^{full}, β_D^{exp}, β_D^{bl}	protocol
h, $T(h)$, $\mathrm{LR}(h)$, M	`docs/57` (local there)
n_{sim}, Pr, $\mathbb{E}$	canonical
$\ell_1(h)$, ℓ_0	`docs/57` notation, redefined here (ℓ_0 is the model without any drug term)
σ_u^2, ρ_a	canonical
Λ	canonical (noncentrality)
$p(i)$, $\mathcal{T}$, r_1, μ_p, μ^0, Σ_p, Σ, $\bar{\Sigma}$	defined here
$X_p(h)$, $\mathbf{x}_p(h)$, Z, ξ, ξ^*, V, $\hat{V}_0$, B	defined here
$\omega_p(h)$, $\bar{\omega}$, $b_0(h)$, $\bar{\Psi}_\omega(h; t_{1/2})$, $\bar{d}(h)$, $\mathbf{r}_p$	defined here
$\hat{\beta}_D^{\mathrm{exp}}(h)$, $\hat{C}$, $\hat{K}$, $c_{\max}$, $\mathcal{V}$, $U_i(h)$, h^*, I_{hh}	defined here
$\mathbf{L}$, $\mathbf{L}^*$ (label vectors), G, $\mathcal{S}(\cdot)$ (a statistic)	defined here
ς	defined here (Berger-Boos allowance; γ is reserved)
$z_{0.975}$, $t_{0.975}$, Φ, χ^2_1	standard

(Notation, 2026-10-09: R is now n_{sim}, $u_i(h)$ is now $U_i(h)$, probability is Pr and expectation $\mathbb{E}$, following `NOTATION.md`; entries that `NOTATION.md` now lists are marked canonical. The unit drug-response profile, written $\mathbf{q}_p$ and q_{pt} in the protocol and here until 2026-10-09, is now Ψ_p and Ψ_{pt} in both, because q is the canonical path index; Ψ is added to `NOTATION.md`. The protocol's path letter p is kept.)

Reserved symbols checked: γ, δ, κ, ν, θ and λ are not used; π is used only for power. `perl tools/notation-lint.pl` reports the file clean (Section 15).